

BCSE205L

Computer Architecture and Organization

Module 6 – Reliability of Memory Systems

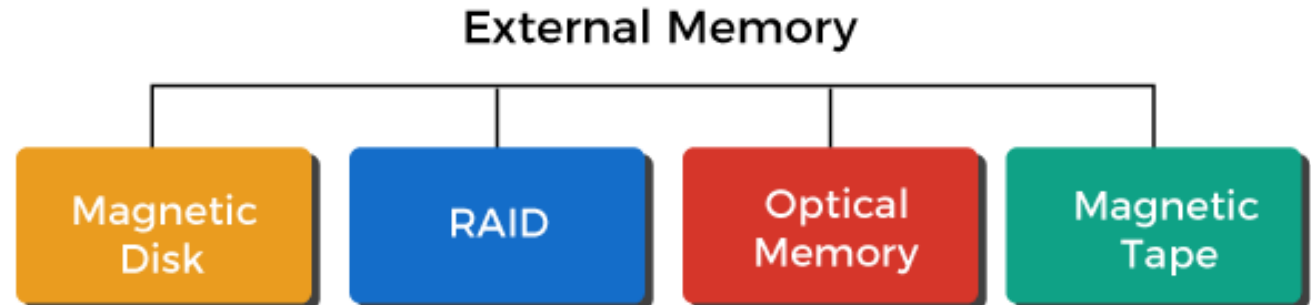
Module:6	Subsystems	5 Hours
External storage systems: Solid state drivers - Organization and Structure of disk drives: Electronic- magnetic and optical technologies - Reliability of memory systems - Error detecting and error correcting systems - RAID Levels - I/O Performance		

External Storage Systems

- Why secondary storage?
 - main memory too small
 - main memory volatile
- Devices: Disks, Tapes, Drums
- Disks are critical components
 - For virtual memory management
 - Storing files
- Disk technology hasn't changed much compared to processor technology (esp., speed)

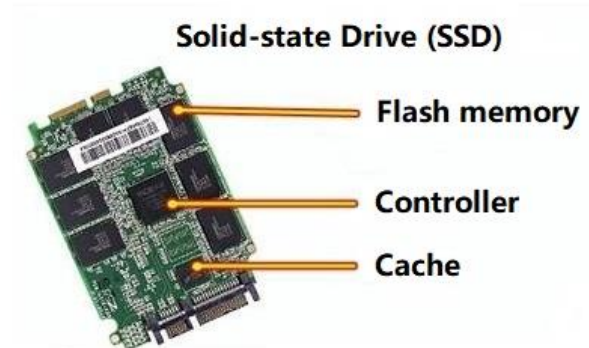
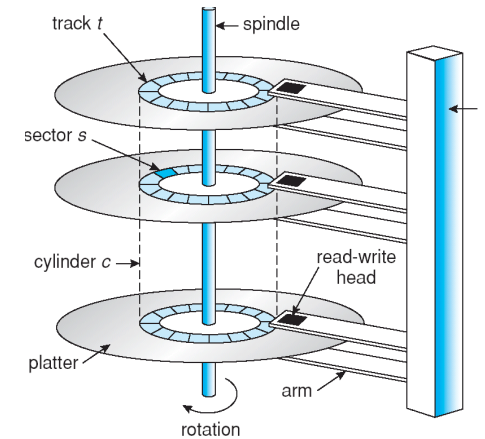
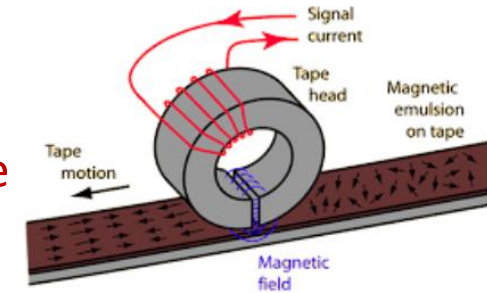
External Storage Systems

- **External memory** - secondary memory or backing store
- Used to store a huge amount of data because it has a **huge capacity**.
- At present, it can measure the data in hundreds of megabytes (MB) / gigabytes (GB) / TB...
- The important property of external memory is that whenever the computer switches off, then **stored information will not be lost**.
- The external memory can be categorized into four types:
 - 1.Magnetic disk
 - 2.Magnetic Tape
 - 3.Optical memory
 - 4.RAID



External Storage Systems - Examples

- Magnetic tape
 - An external storage medium that consists of a **loop of flexible celluloid-like material** that can store data in the form of tiny magnetic fields.
- Magnetic disk
 - A storage device that **uses magnetization** to write, rewrite, and access data. The disk is divided into **platters**, which are made up of concentric circles called **tracks**.
- Hard disk drive (HDD)
 - A **larger-capacity** external storage unit that plugs into a computer or console for long-term storage. Hard drives are ideal for storing files and applications because of their size, capacity, and stability
- Solid state drive (SSD)
 - A **non-volatile data storage device** that can be used **internally or externally**. SSDs store data on **microchips or integrated solid-state flash memory** and **don't have moving parts**.



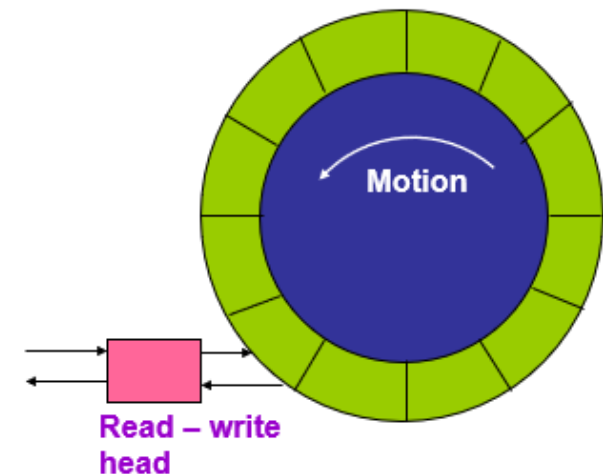
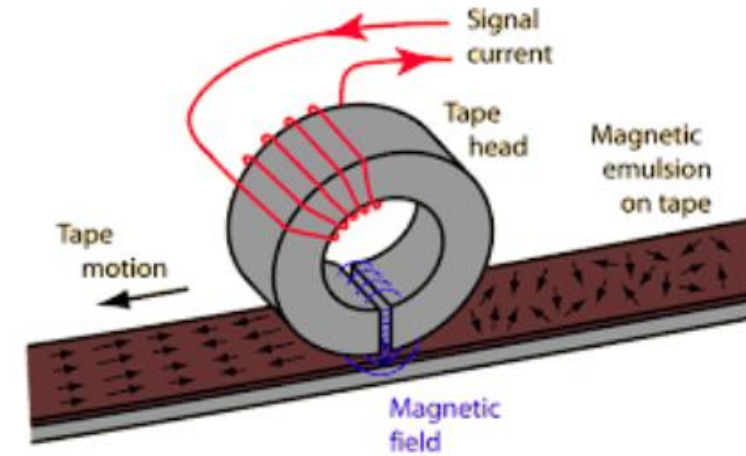
External Storage Systems -Magnetic Technologies

Magnetic Technologies:

Magnetic drums, magnetic tape and magnetic disks are types of magnetic memory. These memories use property of magnetic memory.

1.Magnetic Tape

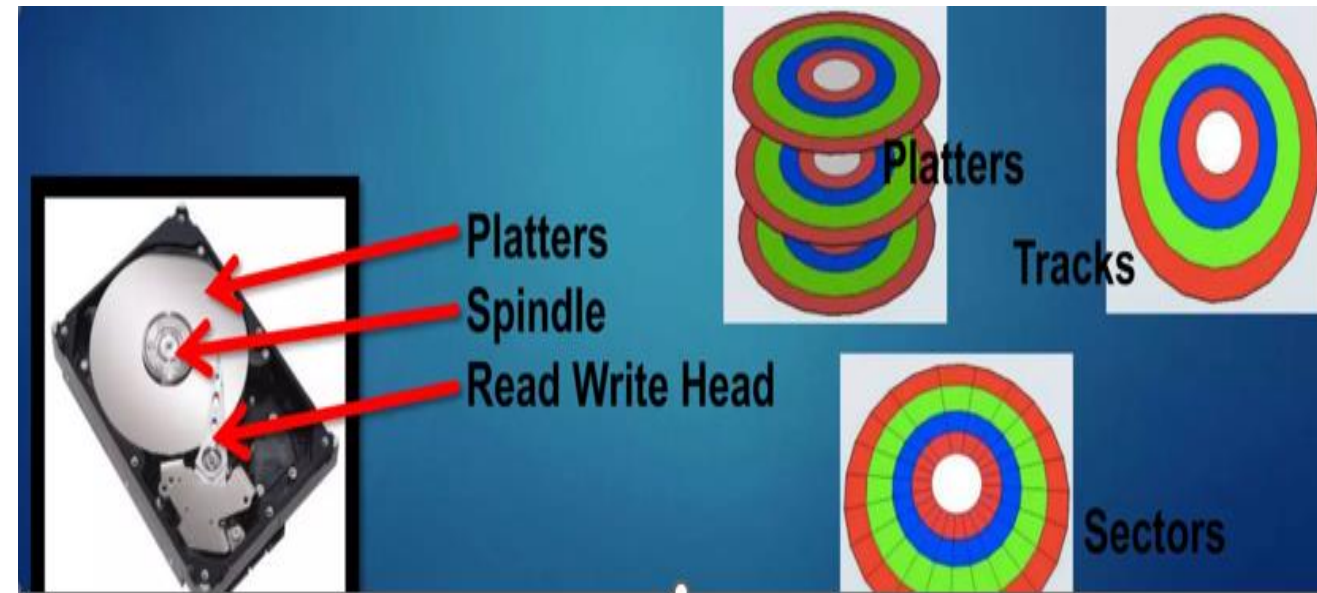
- In **magnetic tape only one side of the ribbon** is used for storing data.
- It is **sequential memory** which contains thin plastic ribbon to store data and coated by magnetic oxide.
- Data read/write speed is **slower** because of sequential access.
- It is **highly reliable** - requires magnetic tape drive writing and reading data.



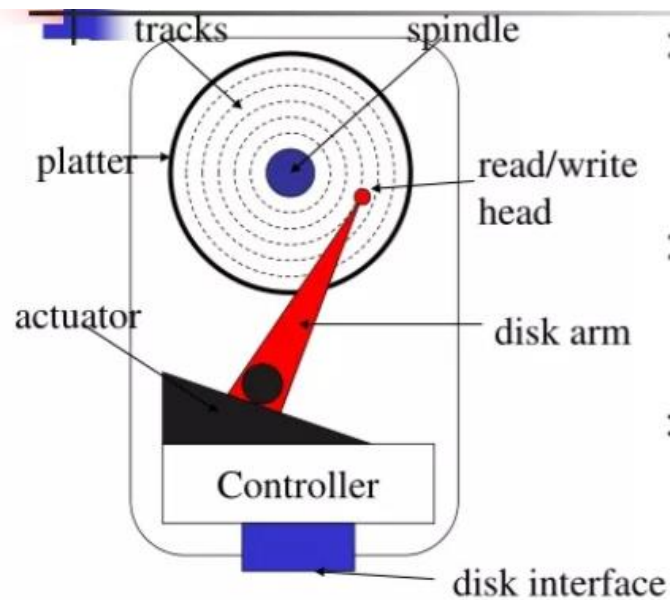
External Storage Systems-Magnetic Technologies

2.Magnetic Disk

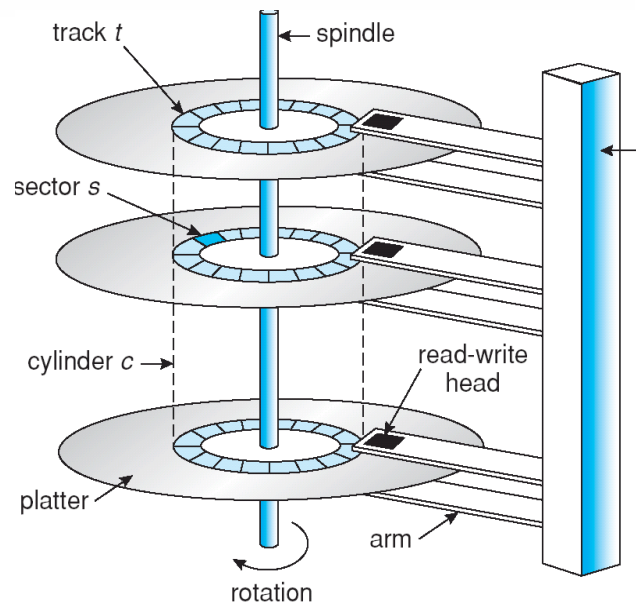
- A magnetic disk is a storage device that **uses a magnetization process** to read, write, rewrite and access data.
- A type of secondary memory that is a **flat disc covered with a magnetic coating to hold information.**
- The Magnetic disk is made of a **set of circular platters.**
 - It is covered with a magnetic coating and stores data in the form of **tracks, spots, and sectors.**
- **Less expensive** than RAM
- Can store **large amounts of data**
- **Data access rate is slower** than main memory because of secondary memory.
- Data can be **modified or can be deleted easily** in the magnetic disk memory.
- It also **allows random access** to data.



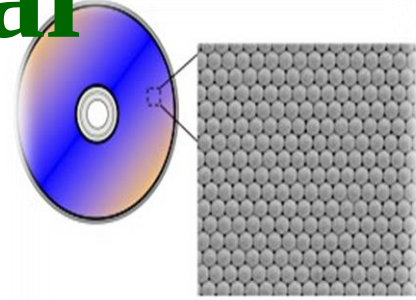
External Storage Systems-Magnetic Technologies



- A **Hard disk Drive (HDD)** includes two main elements =>
 - a spinning platter
 - an actuator arm.
- The **platter** is a circular magnetic disk containing tracks and sectors that retain data.
 - **“Platter”** – Circular Piece of Magnetic Material – Coated Aluminium
 - One Platter Has **2 Sides** - Each Side is Divided into Concentric Rings Called **“Tracks”** - Each Track is Divided into Arcs Called **“Sectors”**.
 - The standard size of each sector is 512Bytes (Note: 512 bytes is a universal size)
- The **actuator arm** moves across the platter to read and write data.



External Storage Systems-Optical Technologies



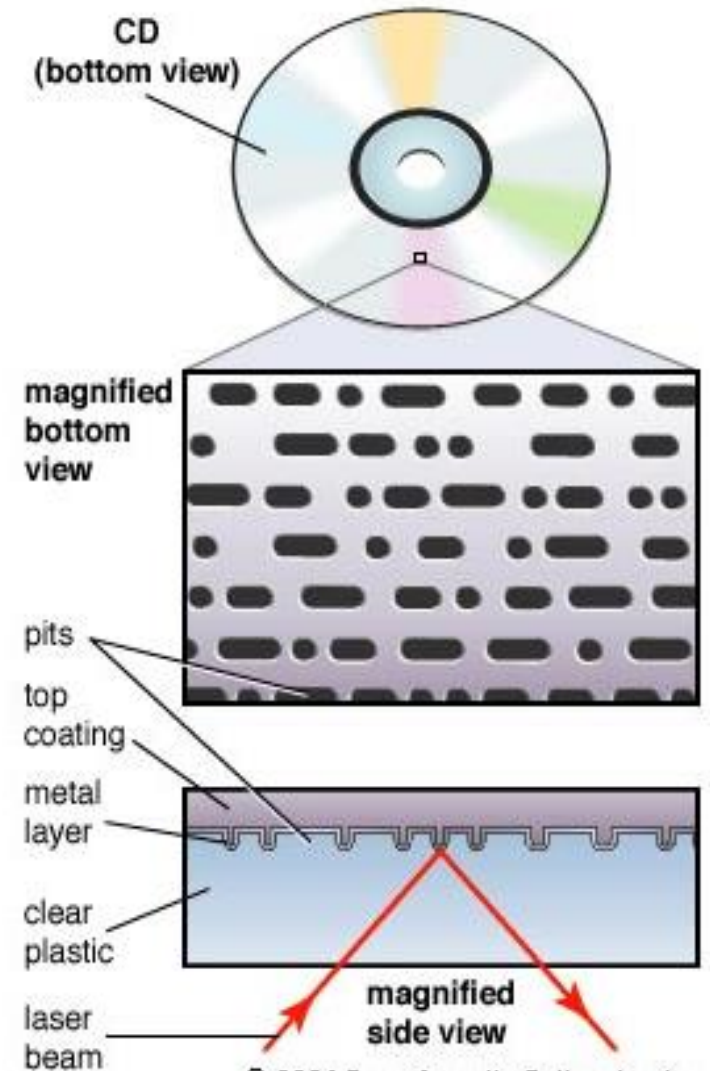
3. Optical Disc

- The Optical memory is an **electronic storage medium** that **uses a laser beam to store and retrieve digital (binary) data**.
- In optical storage technology, a **laser beam encodes digital data on an optical disc** or laser disc in the form of tiny pits arranged in a spiral pattern on the surface of the disc.
- also referred to as an optical disk, optical storage, optical media, Optical disc drive, disc drive, **which reads and writes data by using optical storage techniques and technology**.
- **Examples:** Digital Versatile Disk (DVD), Compact Disk (CD) and Blu-Ray Disk

External Storage Systems-Optical Technologies

How does an optical drive work?

- The **spindle** spins the **optical media** (CD, DVD, Blu-ray)
- A **laser** shines through a lens and reflects off the **bottom** of an optical disc
- If the **laser shines in a pit**, the computer reads a **“zero”**
- If the **laser reflects off a land**, the computer reads a **“one”**



External Storage Systems-Solid State Drivers (SSD)

- Solid - State Drive (SSD) is a solid-state storage device that uses integrated circuit assemblies as memory to store data.
- SSD is also known as a solid-state disk although SSDs do not have physical disks.
- There are no moving mechanical components in SSD.
- This makes them different from conventional electromechanical drives such as Hard Disk Drives (HDDs) or floppy disks, which contain movable read/write heads and spinning disks.
- It is a type of non-volatile memory that retains data even when power is lost.

External Storage Systems-Solid State Drivers (SSD)

Classification of SSD:

1. Flash-based solid-state drives

- Flash-based solid-state drives use FLASH chips as storage media, which is also commonly referred to as SSD. Its appearance can be made into a variety of shapes, such as notebook hard drives, micro hard drives, memory cards, U disks, and other styles.

2. DRAM-based solid-state drives

- DRAM-based solid-state drives used DRAM as a storage medium, and the application range is narrow. It imitates the design of a traditional hard disk, which can be used for volume setting and management by most operating system file system tools, and provides industry-standard PCI and FC interfaces for connecting to a host or server.

External Storage Systems-Solid State Drivers (SSD)

Architecture of SSD:

Three components: Controller, Cache, Flash Memory

1.Controller

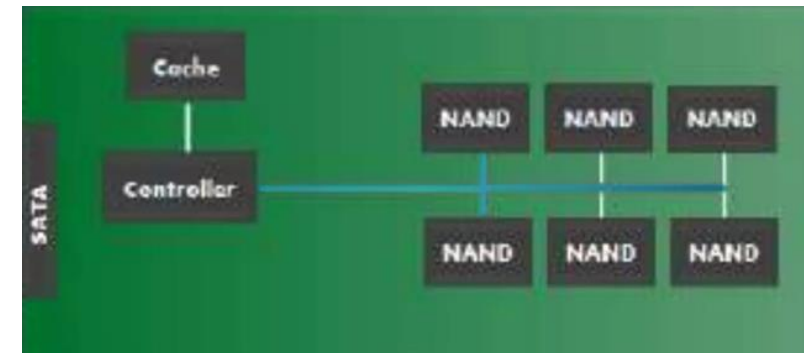
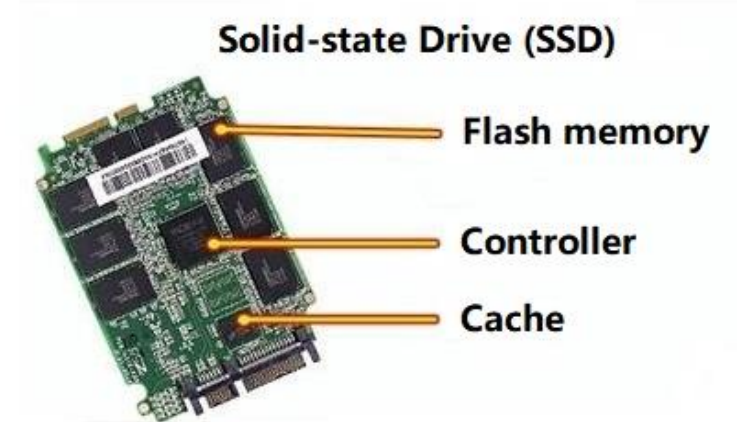
The controller is the **brain of the solid-state drive**. One of its functions is to rationally **allocate the data load on each flash memory**, and the other is to undertake the entire data transfer, connecting the flash memory chip and the external SATA interface.

2.Cache

Next to the controller chip are cache particles. Like traditional hard drives, solid-state drives require a **high-speed cache chip to assist** the main controller for data processing.

3.Flash memory

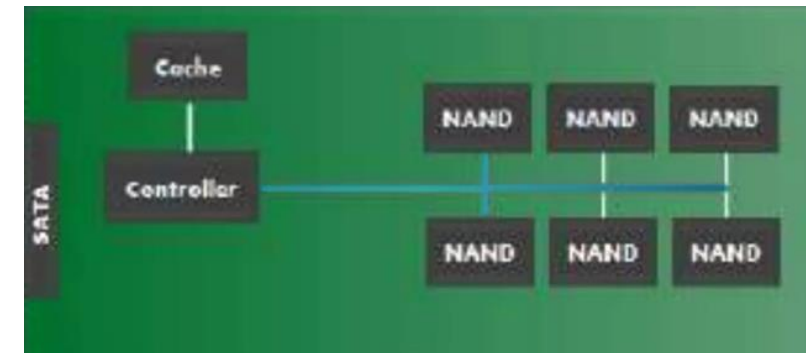
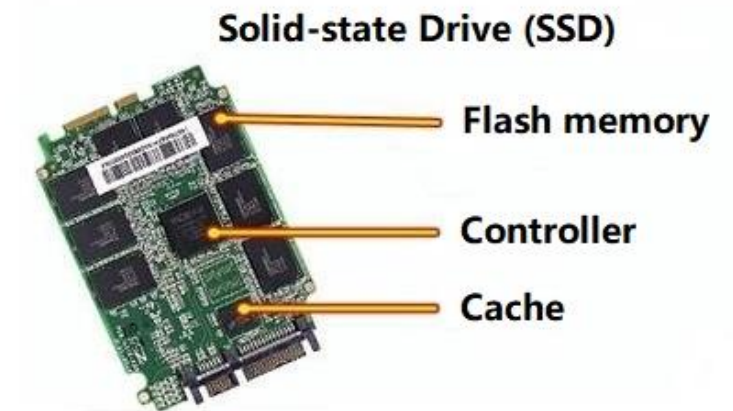
Except for the controller and the cache, most of the other positions on the PCB board are **NAND flash memory chips**.



External Storage Systems-Solid State Drivers (SSD)

Advantages of SSD:

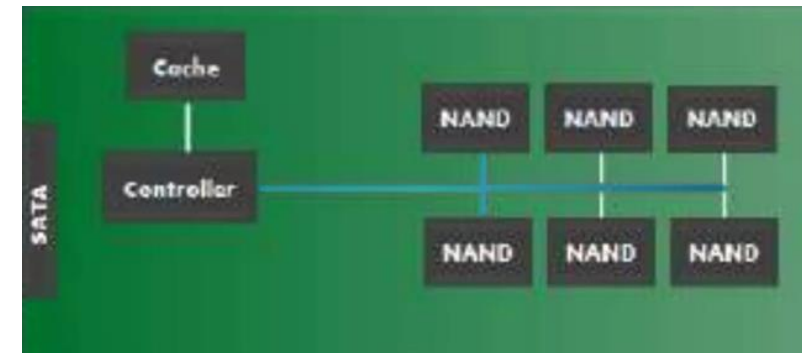
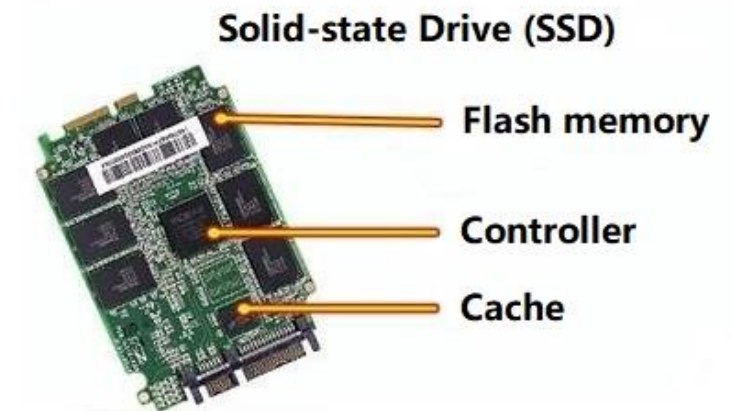
- **Faster read/write speeds**. SSDs can access **large files quickly**.
- **Quicker boot times** and **better performance**. Because the drive does not need to spin up as an HDD would, it is more responsive and provides better load performance.
- **Durability** - SSDs are more shock-resistant and can handle heat better than HDDs because they do not have moving parts.
- **Power consumption** - SSDs need less power to operate than HDDs due to their lack of moving parts.
- **Quieter** - SSDs produce less audible noise because there are no moving or spinning parts.
- **Size** - SSDs come in a variety of form factors whereas HDD sizes are limited.



External Storage Systems-Solid State Drivers (SSD)

Disadvantages of SSD:

- **Cost** - SSDs are more expensive than traditional HDDs.
- **Life expectancy** - Some SSDs, for example, those using NAND memory-flash chips, can only be written a specified number of times that is typically less than HDDs.
- **Performance** - Limitations on the number of **write cycles** cause SSDs to decrease in performance over time.
- **Storage options** - Because of cost, SSDs are **typically sold in smaller sizes**.
- **Data recovery** - This time-consuming process can be expensive, as the data on **damaged chips may not be recoverable**.



External Storage Systems-Solid State Drivers (SSD)

SSD Vs HDD

Solid-State Drive	Hard Disk Drive
Random access time 0.1 ms.	Random access time 5-10 ms.
Read latency time very low.	Read latency time high.
100MB/s to 500MB/s.	50MB/s to 100MB/s.
High Reliability.	Low reliability.
SSDs have no moving parts to fail mechanically.	HDDs have moving parts and are subject to sudden failure.
Small and light weight.	Relatively large and heavy.
Till now SSDs are available in size up to 2TB.	Till now HDDs are available up to 4TB .
Power Consumption 2watts	6-12 watts.

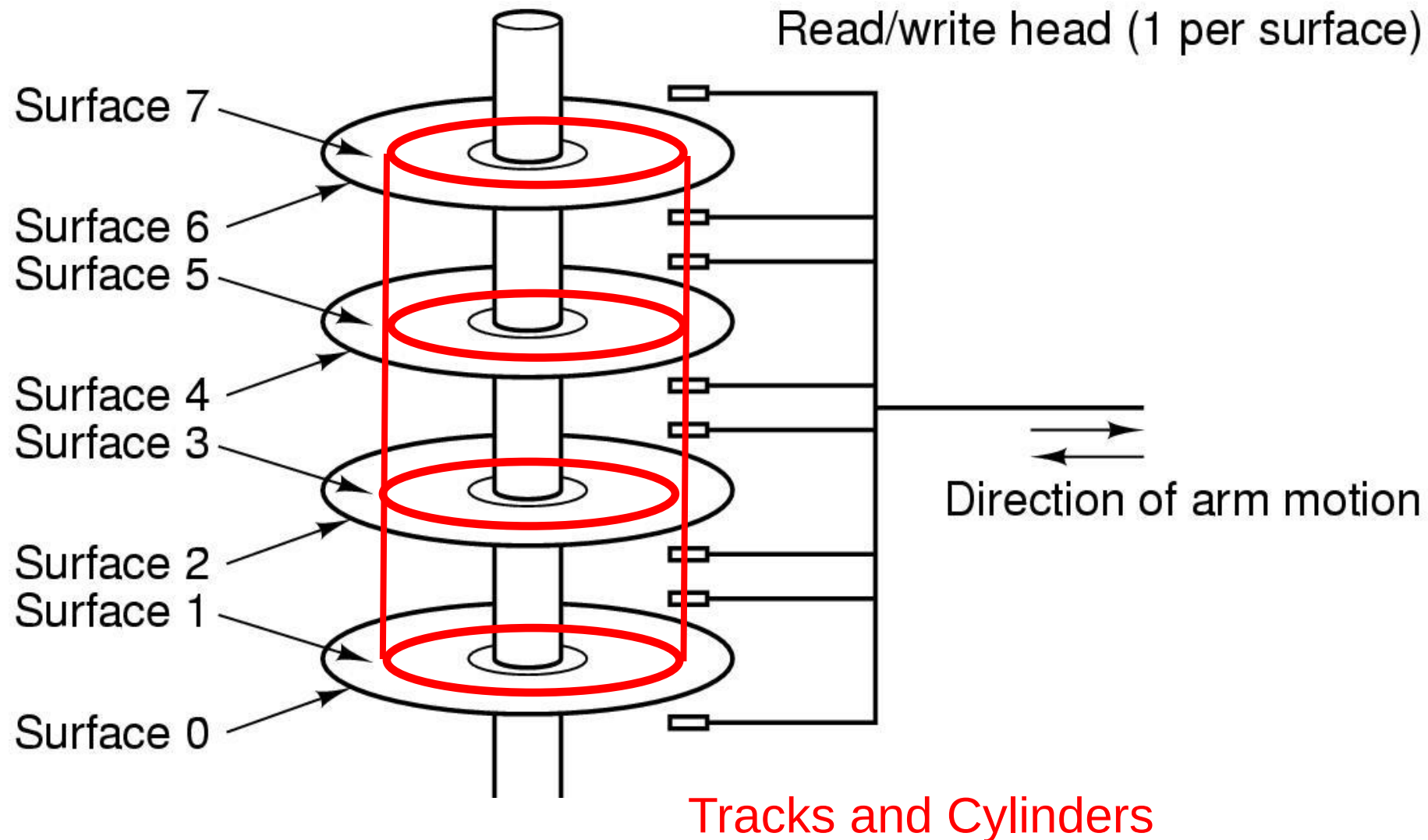
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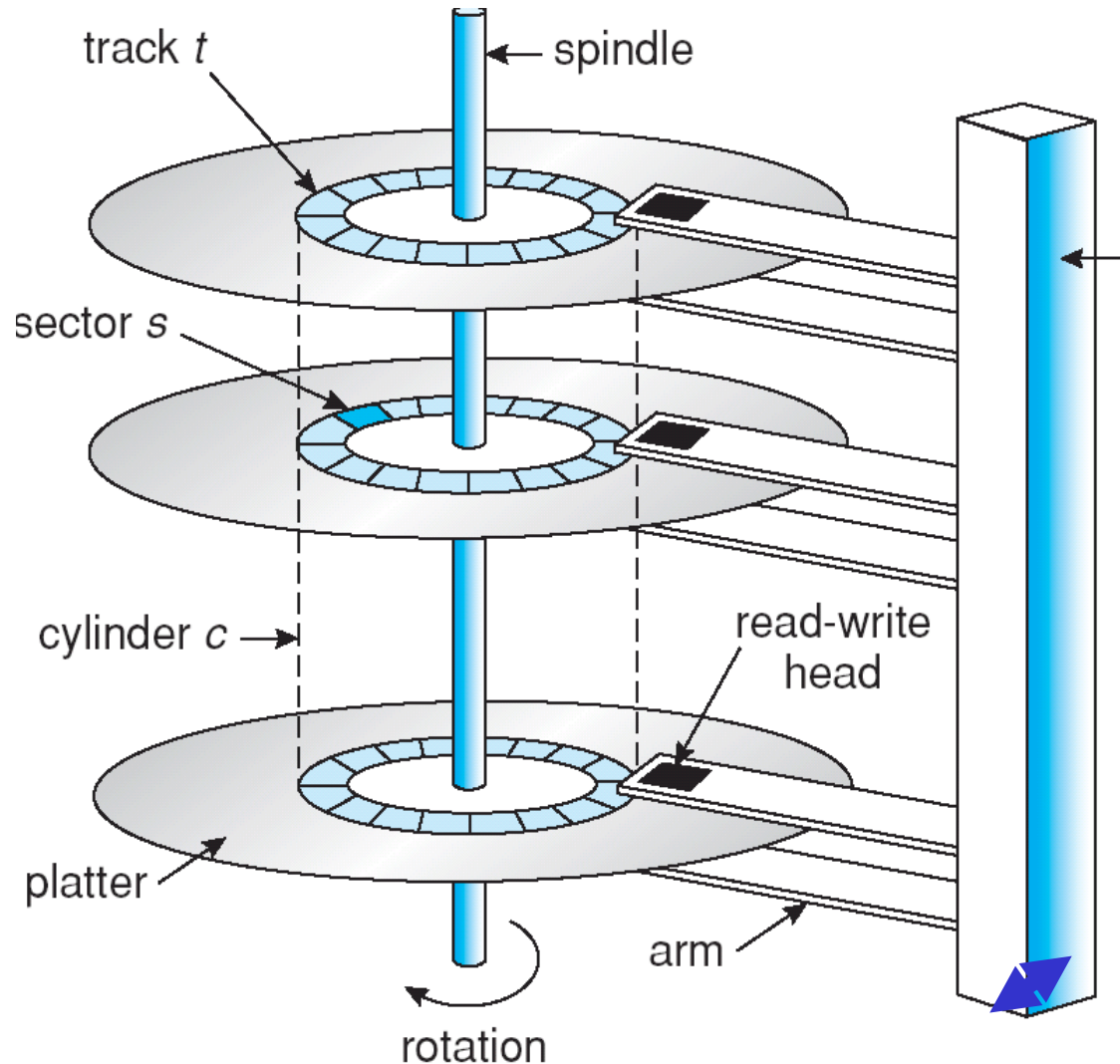
Module 6 - Organization and Structure of Disk Drives

Module:6	Subsystems	5 Hours
External storage systems: Solid state drivers - Organization and Structure of disk drives: Electronic- magnetic and optical technologies - Reliability of memory systems - Error detecting and error correcting systems - RAID Levels - I/O Performance		

Organization and Structure of Disk Drives



Organization and Structure of Disk Drives



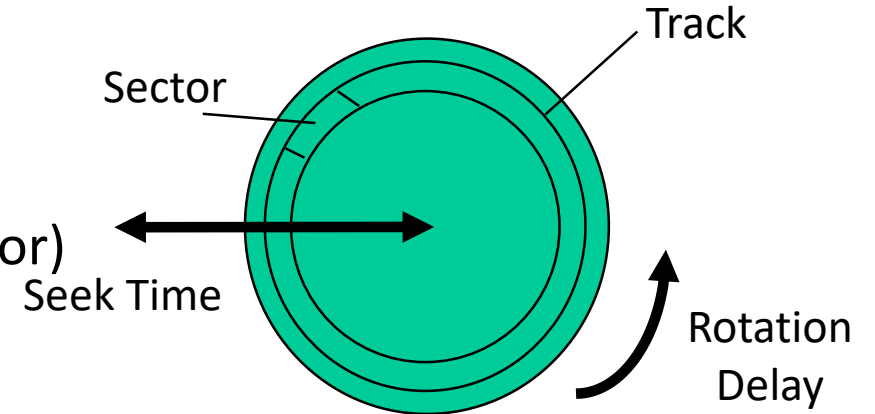
Typical parameters :

- 1 spindle
- 1 arm assembly
- 1-2 sides/platter
- 1 head per side
(but only 1 active head at a time)
- 700-20480 tracks/surface
- 16-1600 sectors/track

Organization and Structure of Disk Drives

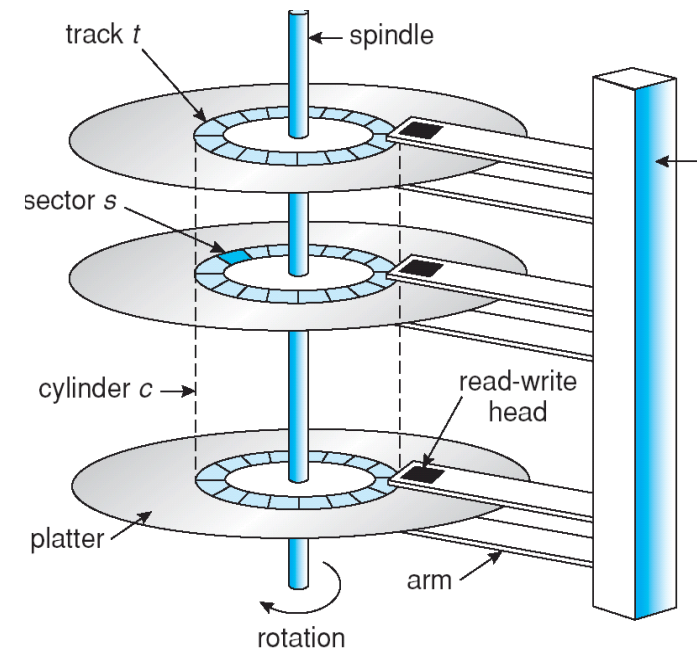
- **Accessing a disk requires:**

- specify sector: C (cylinder), H (head), and S (sector)
- specify size: number of sectors to read or write
- specify memory address



- **Performance:**

- Seek time: move the arm assembly to track
- Rotational delay: wait for sector to come around
- Transfer time: get the bits off the disk
- Controller time: time for setup



Organization and Structure of Disk Drives

Typical Disk Parameters

Parameter	IBM 360-KB floppy disk	WD 18300 hard disk
Number of cylinders	40	10601
Tracks per cylinder	2	12
Sectors per track	9	281 (avg)
Sectors per disk	720	35742000
Bytes per sector	512	512
Disk capacity	360 KB	18.3 GB
Seek time (adjacent cylinders)	6 msec	0.8 msec
Seek time (average case)	77 msec	6.9 msec
Rotation time	200 msec	8.33 msec
Motor stop/start time	250 msec	20 sec
Time to transfer 1 sector	22 msec	17 μ sec

Compare Seek time, rotation time, transfer time

Organization and Structure of Disk Drives

- Disk Access

Design considerations

- Seek time (moving head to correct cylinder) and rotational latency time (waiting for correct sector) are **greater than** data transfer time (time to read)
- Lots of errors possible (bad sectors, bad bits)
- **Caching blocks** that happen to pass under disk head **commonly used**

Average time = Seek time + rotational delay + transfer time + controller overhead

Organization and Structure of Disk Drives

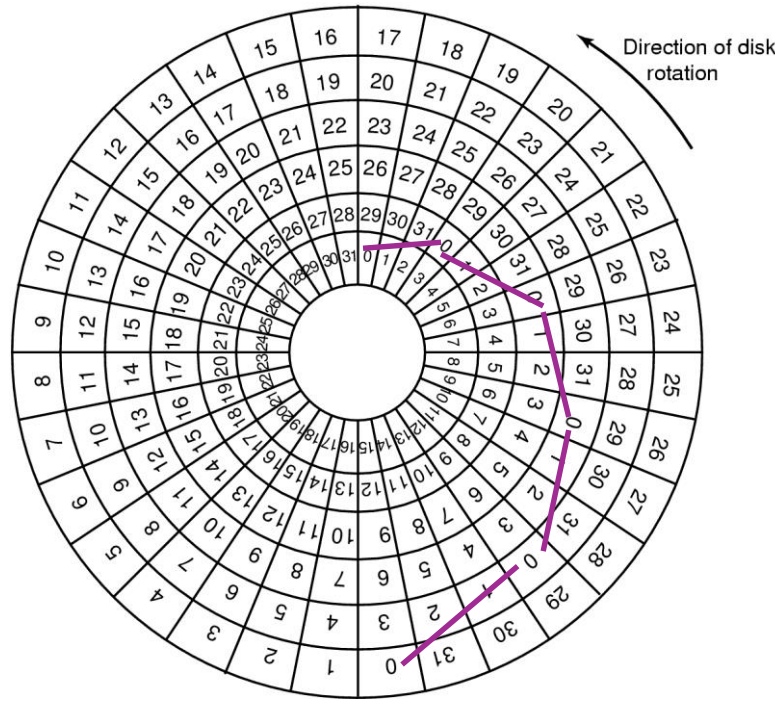
- **Disk Format**
- Typical sector (512 bytes)
 - **Preamble** identifying start code and sector address
 - **Data**
 - **Error correction code (16 bits)** - At least detecting errors possible with probability almost 1



A disk sector

Organization and Structure of Disk Drives

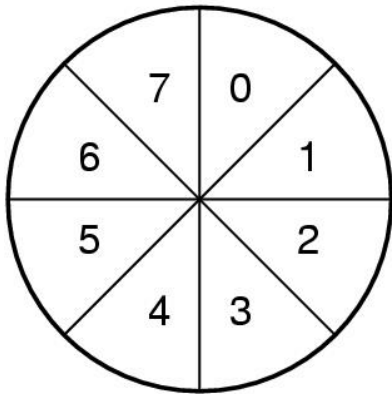
- Disk Format
- Cylinder skew
 - After reading an entire track, as head is moved across a cylinder, seeking next sector shouldn't cause waiting for full rotation



An illustration
of cylinder skew

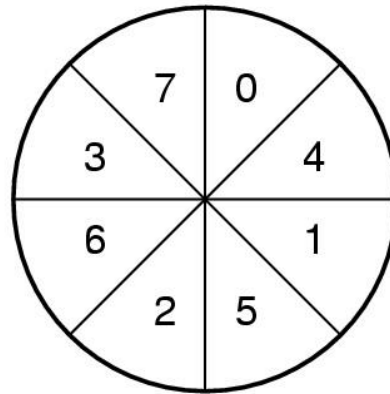
Organization and Structure of Disk Drives

- **Disk Format**
- Interleaving sectors
 - As a sector is copied into controller buffer, it needs to be copied into main memory. So next sector should not be adjacent if we wish to avoid waiting for one full rotation



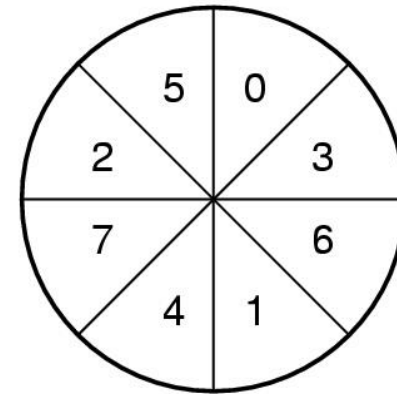
(a)

No interleaving



(b)

Single interleaving



(c)

Double interleaving

Organization and Structure of Disk Drives

- **Disk Scheduling – I/O scheduling**

- Time required to read or write a disk block determined by **3 factors**
 - **Seek time** (moving head to correct cylinder)
 - **Rotational delay** (waiting for correct sector)
 - **Actual transfer time** (data transfer time)
- Seek time dominates
- Error checking is done by controllers
- **Disk Scheduling** - schedule I/O requests arriving for the disk

- **Disk Scheduling Algorithms/Techniques:**

- **FCFS (First Come First Serve)**
- **SSTF (Shortest Seek-Time First)**
- **SCAN (Elevator Algorithm)**
- **C-SCAN (Circular SCAN)**
- **LOOK**
- **C-LOOK**

Organization and Structure of Disk Drives

- **Key Terms Associated with Disk Scheduling**

- **Seek Time:** Seek time is the time taken to locate the disk arm to a specified track where the data is to be read or written. So the disk scheduling algorithm that gives a minimum average seek time is better.
- **Rotational Latency:** Rotational Latency is the time taken by the desired sector of the disk to rotate into a position so that it can access the read/write heads. So the disk scheduling algorithm that gives minimum rotational latency is better.
- **Transfer Time:** Transfer time is the time to transfer the data. It depends on the rotating speed of the disk and the number of bytes to be transferred.

- **Disk Access Time:**

$$\text{Disk Access Time} = \text{Seek Time} + \text{Rotational Latency} + \text{Transfer Time}$$

- **Total Seek Time:**

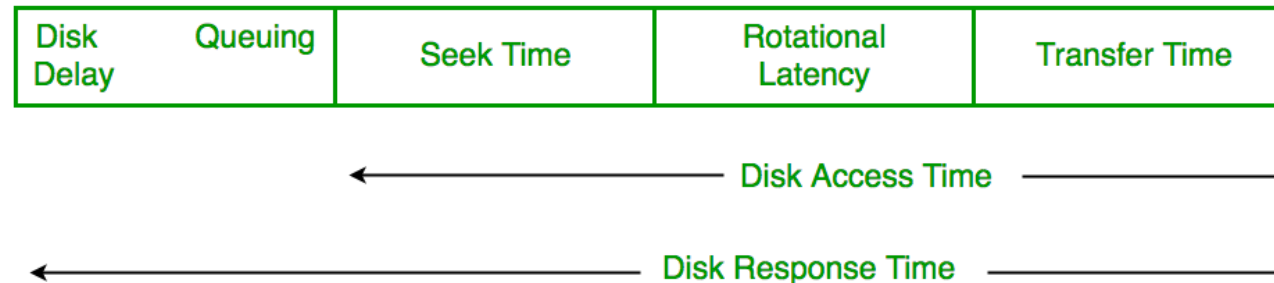
$$\text{Total Seek Time} = \text{Total head Movement} * \text{Seek Time}.$$

Organization and Structure of Disk Drives

- Key Terms Associated with Disk Scheduling

- Disk Access Time:

Disk Access Time = Seek Time + Rotational Latency + Transfer Time



- **Disk Response Time:** Response Time is the average time spent by a request waiting to perform its I/O operation.
 - **Average Response time** is the response time of all requests.
 - **Variance Response Time** is the measure of how individual requests are serviced with respect to average response time. So the disk scheduling algorithm that gives minimum variance response time is better.

Organization and Structure of Disk Drives

- **Disk Scheduling Algorithms/Techniques - FCFS**

- Basic strategy for moving disk arm across cylinders: First-come-first-served queue

Example:

Disk queue: 98, 183, 37, 122, 14, 124, 65, 67 .

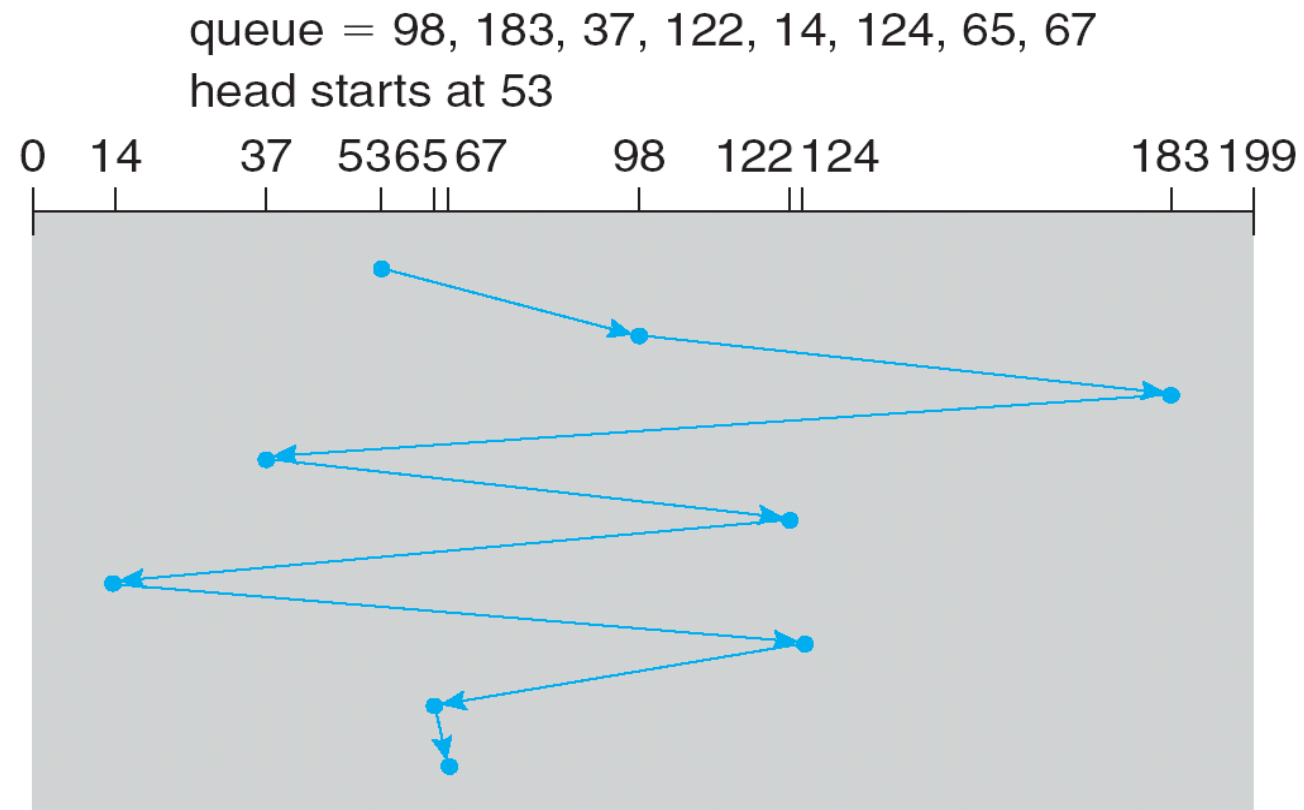
Initial r/w head position: cylinder 53.

Total head movement

$$= 45 + 85 + 146 + 85 + 108 + 110 + 59 + 2$$

$$= 640 \text{ tracks}$$

OK for small load, low variance of waiting time



Organization and Structure of Disk Drives

- **Disk Scheduling Algorithms/Techniques - FCFS**

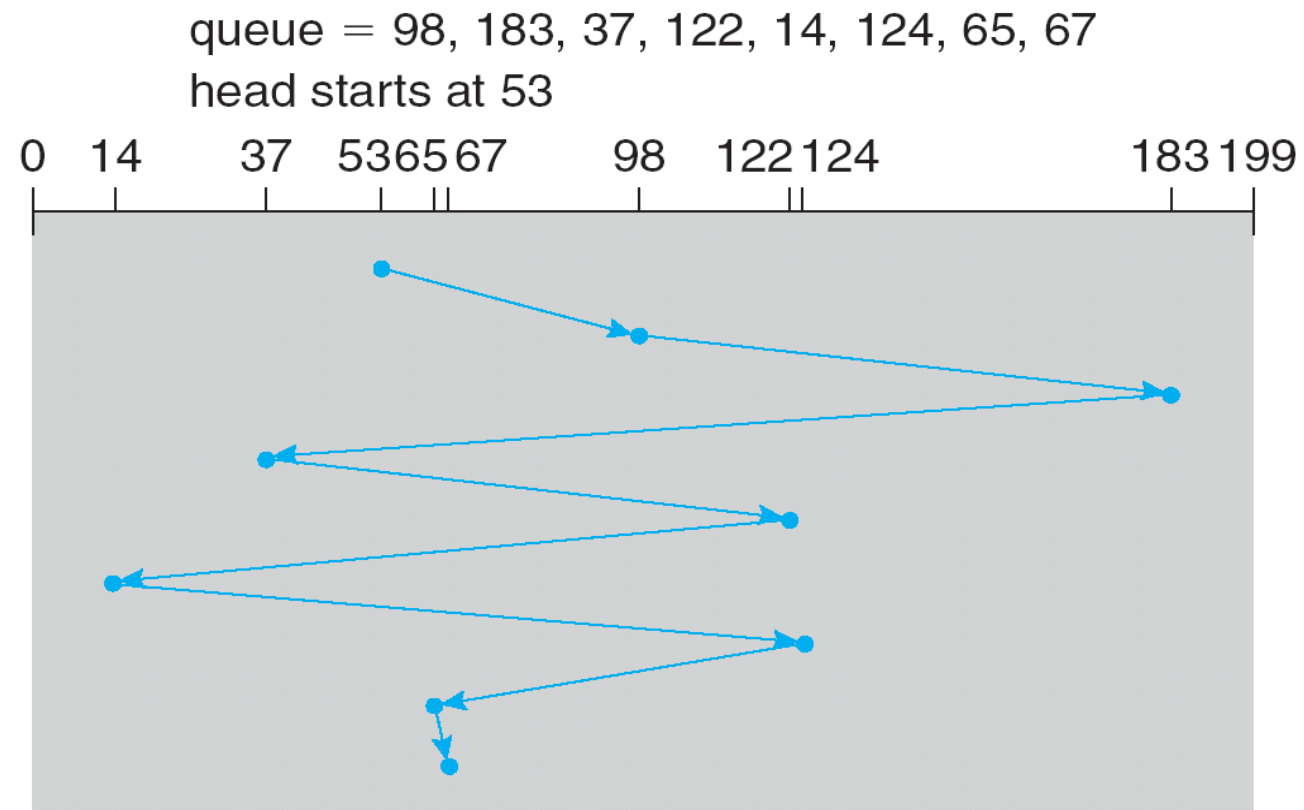
- Basic strategy for moving disk arm across cylinders: First-come-first-served queue

Advantages of FCFS Algorithm

- *Every request gets a fair chance*
- *No indefinite postponement*

Disadvantages of FCFS Algorithm

- *Does not try to optimize seek time*
- *May not provide the best possible service*



Organization and Structure of Disk Drives

- **Disk Scheduling Algorithms/Techniques – SSTF**

- Shortest Seek-Time First (SSTF): Pick the closest one next

Example:

Disk queue: 98, 183, 37, 122, 14, 124, 65, 67

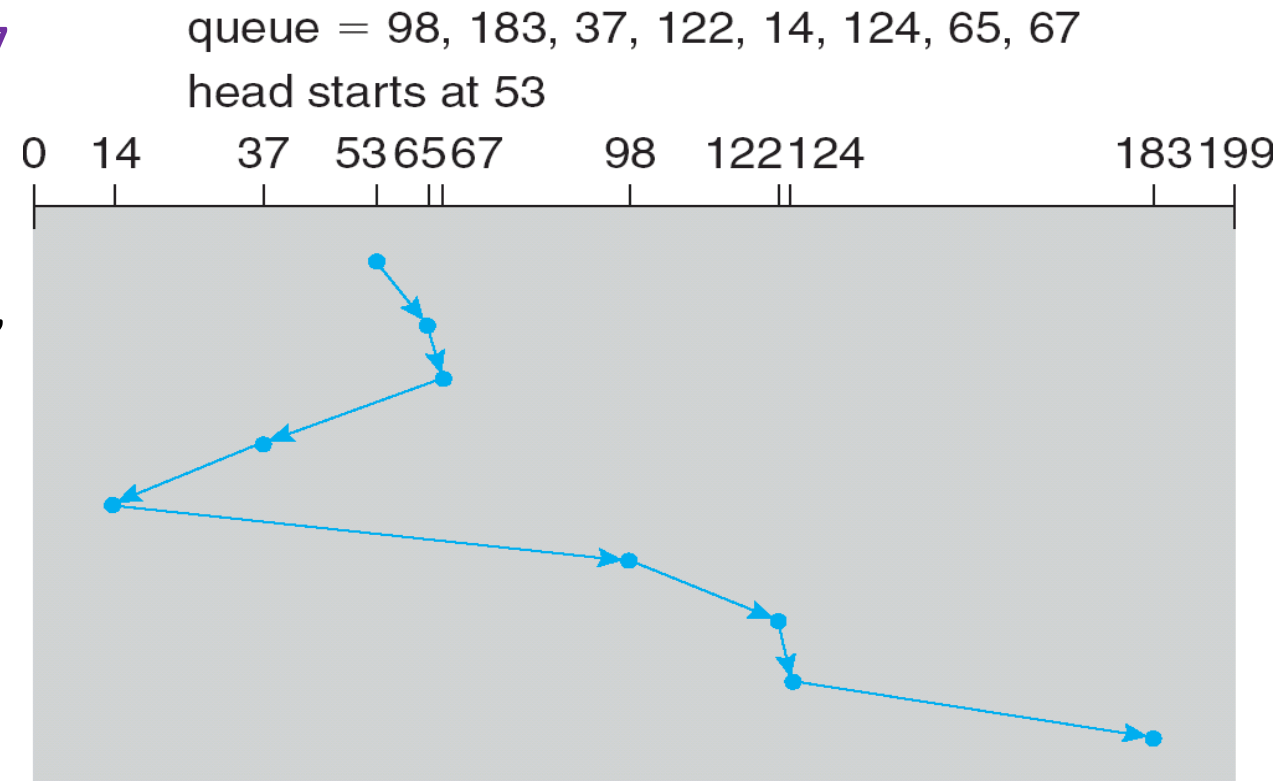
Initial r/w head position: cylinder 53.

Solution:

SSTF ordering will be 65, 67, 37, 14, 98, 122, 124, 183

Total head movement (starting at 53)
= $12 + 2 + 30 + 23 + 84 + 24 + 2 + 59$
= 236 tracks

High variance (starvation possible)



Organization and Structure of Disk Drives

- **Disk Scheduling Algorithms/Techniques – SSTF**

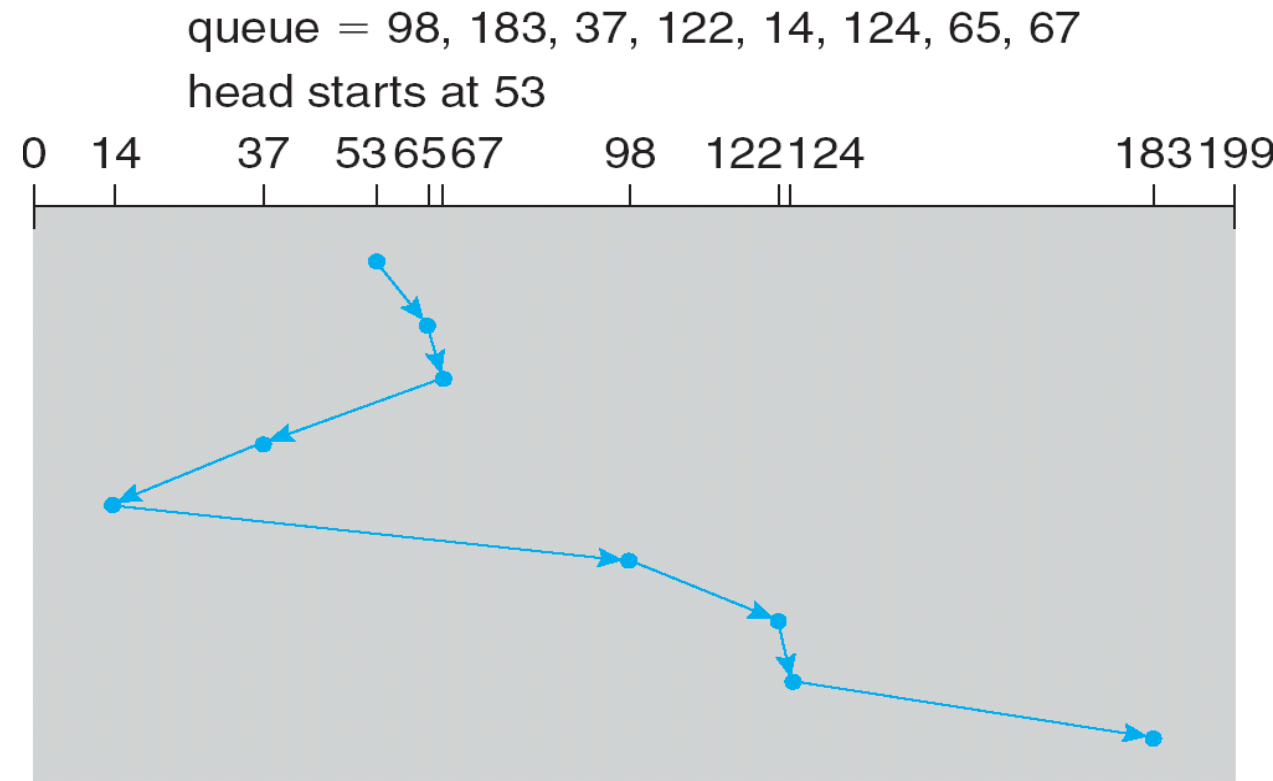
- Shortest Seek-Time First (SSTF): Pick the closest one next

Advantages of SSTF Algorithm

- *The average Response Time decreases*
- *Throughput increases*

Disadvantages of SSTF Algorithm

- *Overhead to calculate seek time in advance*
- *Can cause Starvation for a request if it has a higher seek time as compared to incoming requests*
- *The high variance of response time as SSTF favors only some requests*



Organization and Structure of Disk Drives

- **Disk Scheduling Algorithms/Techniques – SCAN**

- Maintain a direction bit - Pick the request with shortest seek time in current direction first, and change direction only at edges.

Example:

Disk queue: 98, 183, 37, 122, 14, 124, 65, 67

Initial r/w head position: cylinder 53.

Assume head was moving toward 0.

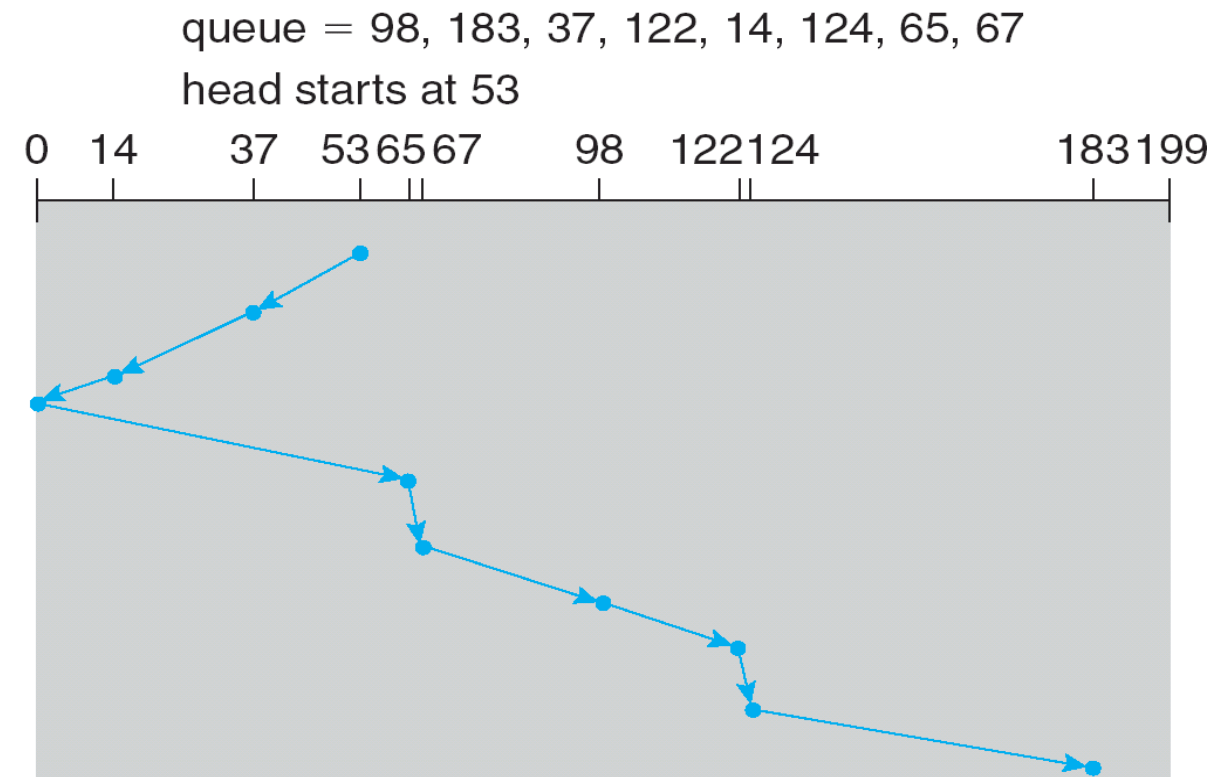
Solution: SCAN ordering will be 37, 14, 0, 65, 67, 98, 122, 124, 183

Total head movement (starting at 53)

$$= 16 + 23 + 14 + 65 + 2 + 31 + 24 + 2 + 59$$

$$= 236 \text{ tracks}$$

Almost as good as SSTF in mean wait time, lower variance.



Organization and Structure of Disk Drives

- **Disk Scheduling Algorithms/Techniques – SCAN**

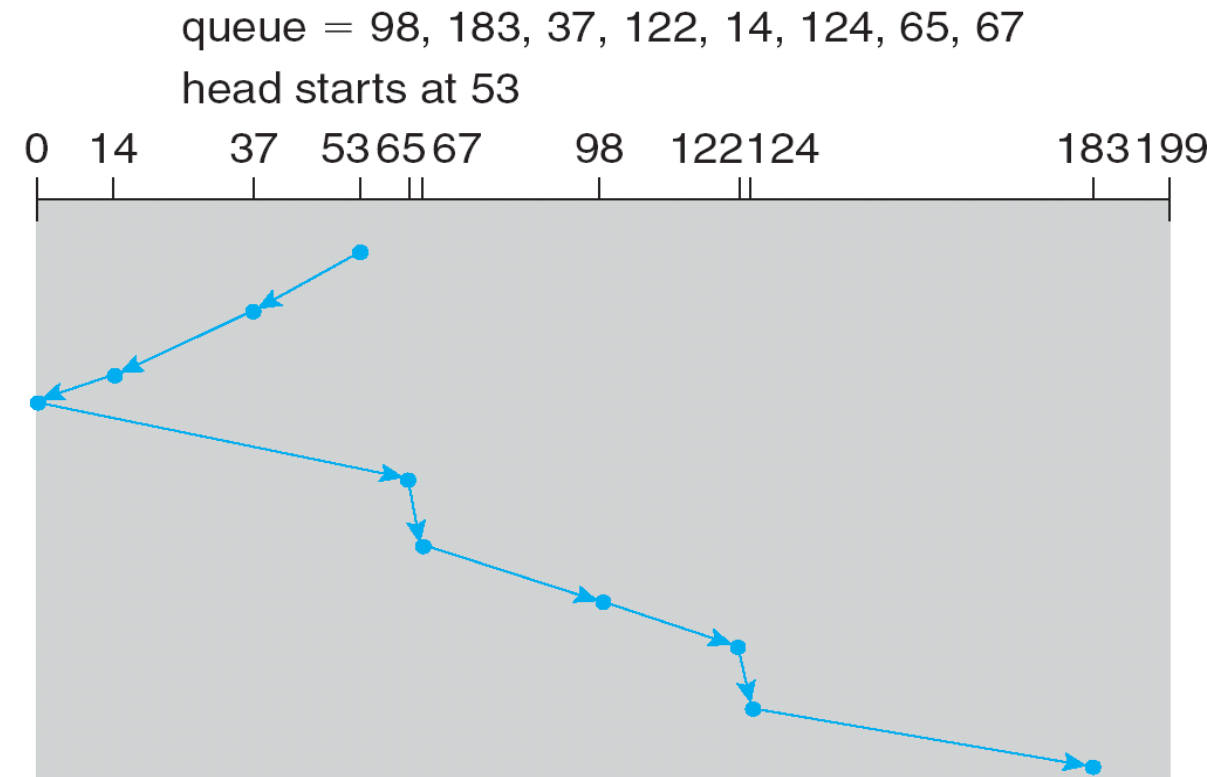
- Maintain a direction bit - Pick the request with shortest seek time in current direction first, and change direction only at edges.

Advantages of SCAN Algorithm

- *High throughput*
- *Low variance of response time*
- *Average response time*

Disadvantages of SCAN Algorithm

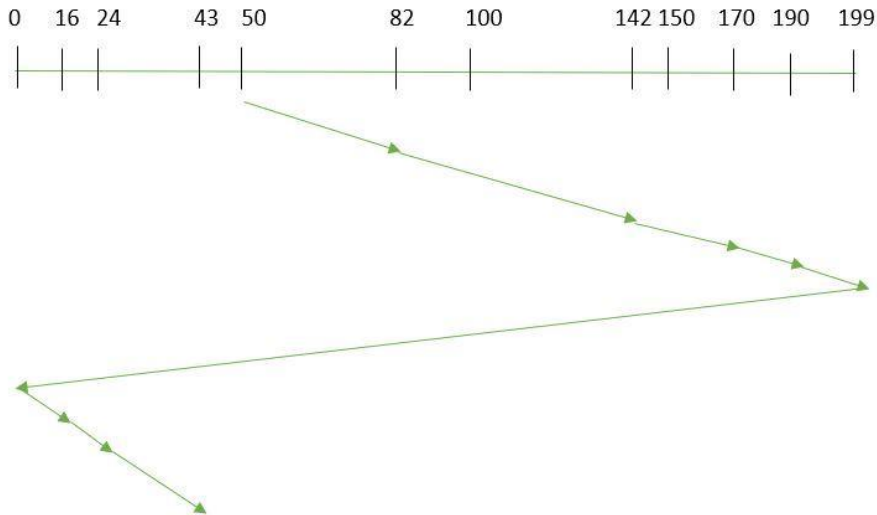
- *Long waiting time for requests for locations just visited by disk arm*



Organization and Structure of Disk Drives

- **Disk Scheduling Algorithms/Techniques – C – SCAN**

- disk arm **instead of reversing its direction goes to the other end of the disk** and starts servicing the requests from there.
- So, the disk arm moves in a **circular fashion** and this algorithm is also similar to the SCAN algorithm hence it is known as **C-SCAN (Circular SCAN)**.

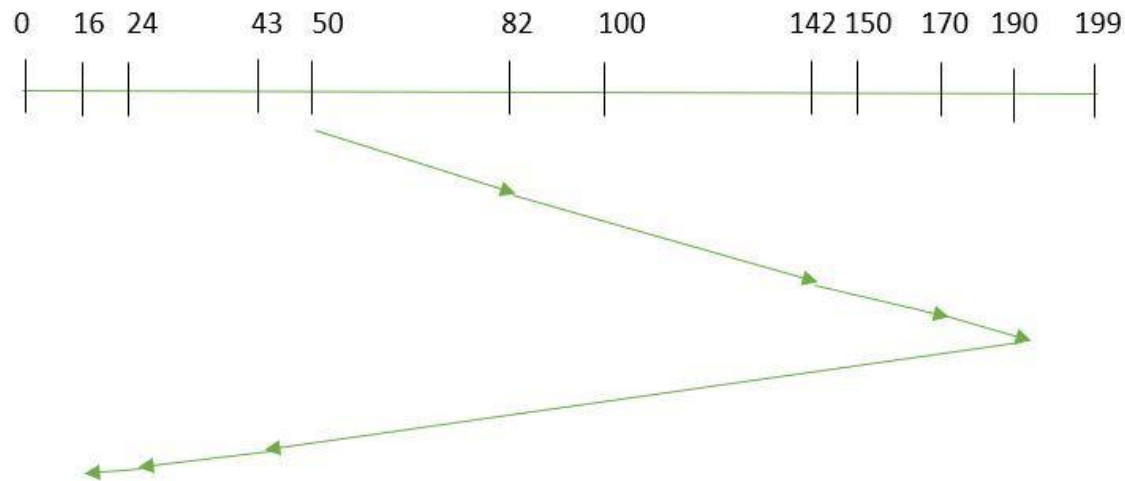


Provides more uniform wait time compared to SCAN.

Organization and Structure of Disk Drives

- **Disk Scheduling Algorithms/Techniques – LOOK**

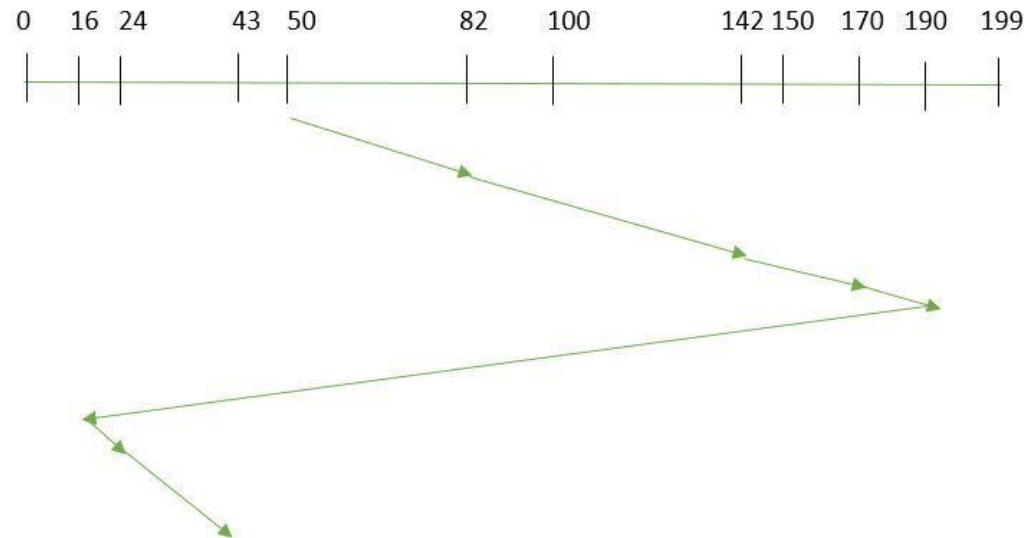
- similar to the SCAN - disk arm in spite of going to the end of the disk goes only to the last request to be serviced in front of the head and then reverses its direction from there only.
- Thus it prevents the extra delay which occurred due to unnecessary traversal to the end of the disk.



Organization and Structure of Disk Drives

- **Disk Scheduling Algorithms/Techniques – C-LOOK**

- similar to the C-SCAN - disk arm in spite of going to the end goes only to the last request to be serviced in front of the head and then from there goes to the other end's last request.
- Thus, it also prevents the extra delay which occurred due to unnecessary traversal to the end of the disk.



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Module 6 – Reliability of Memory Systems

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Reliability of Memory Systems

- **Fault tolerance for Disks**

- Disks can have lots of errors
 - Corrupted bits
 - Corrupted sectors
 - **Manufacturing defects**
 - **Disk crashes**
- Many levels of protection implemented
 - Error correction within sectors
 - Extra sectors within a disk
 - **Mirroring**
 - **RAID (redundancy and parallelism)**

RAID

- Disk access times are still slow compared to CPU/memory
- **Natural solution:** Read concurrently from multiple disks
- **RAID (Redundant Array of Independent Disks)** is a technique that makes use of a **combination of multiple disks** instead of using a single disk for increased
 - performance
 - data redundancy or
 - Both
- **RAID Controller** talks to OS just **like a device driver** and manages a suite of disks to improve reliability and performance
- A RAID system consists of **two or more drives working in parallel**.

RAID

- RAID (redundant array of independent disks) is a way of storing the same data in different places on multiple hard disks or solid-state drives (SSDs) to protect data in the case of a drive failure.
- There are different RAID levels, however, and **not all have the goal of providing redundancy.**
- **RAID arrays** appear to the operating system (OS) as a **single logical drive.**
- RAID employs the **techniques** of **disk mirroring** or **disk striping.**
- **Mirroring** will copy identical data onto more than one drive.
- **Striping partitions** help spread data over multiple disk drives.

RAID

- **Striping**

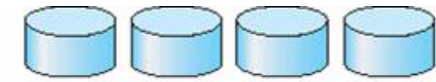
- Organize **logically consecutive chunks of data** across **different disks** to improve performance
- Striping can be at **bit-level, byte-level, or block-level**
- **Incoming request** is split into all relevant disks by **RAID controller**, and their outputs are put together

- **Redundancy through duplication**

- Multiple requests can be **load-balanced**
- **Error recovery** on faults

RAID Levels

- Many different ways of distributing data have been standardized into various RAID levels.
- Each RAID level is offering a trade-off of data protection, system performance, and storage space.
- The number of levels has since expanded and has been broken into **three categories: standard, nested and nonstandard RAID levels.**
- The most popular and standard RAID levels
 - RAID 0
 - RAID 1
 - RAID 2
 - RAID 3
 - RAID 4
 - RAID 5
 - RAID 6



(a) RAID 0: non-redundant striping.



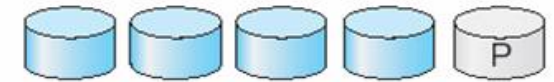
(b) RAID 1: mirrored disks.



(c) RAID 2: memory-style error-correcting codes.



(d) RAID 3: bit-interleaved parity.



(e) RAID 4: block-interleaved parity.



(f) RAID 5: block-interleaved distributed parity.

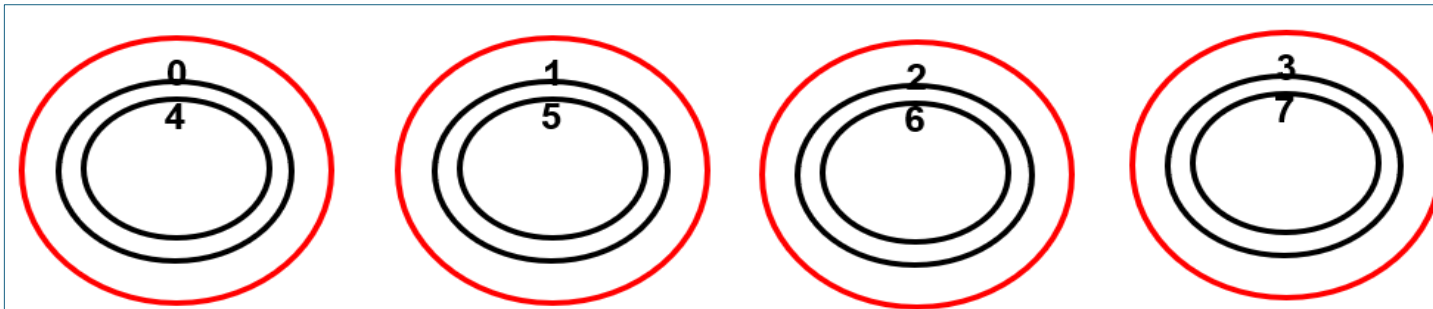


(g) RAID 6: P + Q redundancy.

RAID Levels

RAID Level 0 (Striping without redundancy)

- RAID 0 is primarily focused on improving performance through striping **without providing redundancy** - It offers the best performance, but it does not provide fault tolerance.
- RAID 0 utilizes striping across multiple disks - **Array of disks with block-level striping**
 - Data is divided into **equal-sized blocks (strips)** and distributed in a **round-robin** fashion across all the disks in the array.
 - Strip = k sectors (for suitably chosen k)
 - First k sectors are on first strip of first disk, next k sectors are on first strip of second disk, and so on

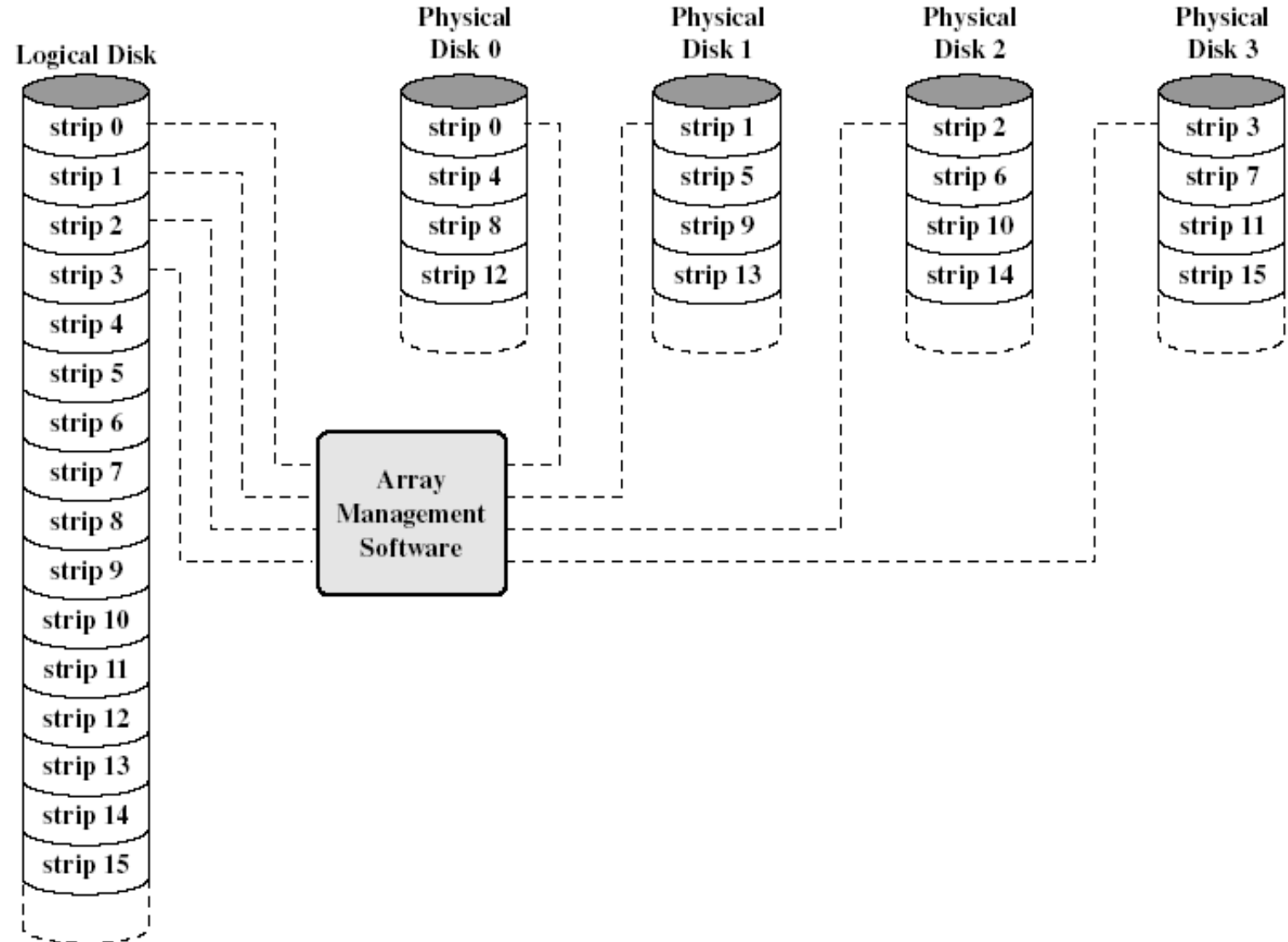


RAID Levels

RAID Level 0 – Data

Mapping

This software may execute either in the disk sub system or in a host computer



RAID Levels

RAID Level 0 – Advantages

- A large request spanning multiple strips can be parallelized easily
- Not much overhead
- Used in high-performance applications

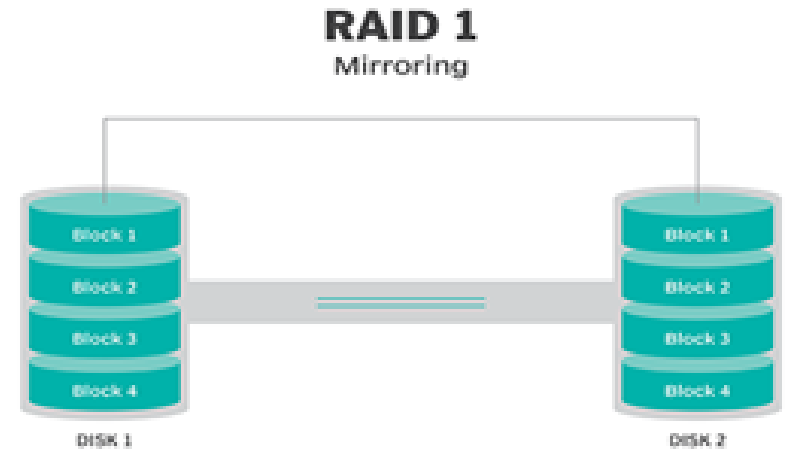
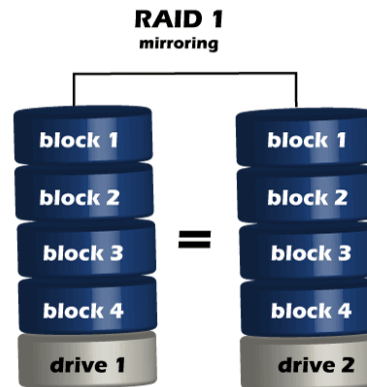
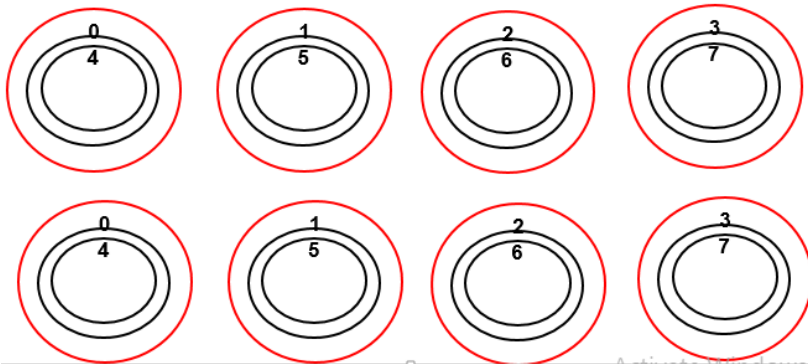
Disadvantages

- No redundancy
 - This is the biggest disadvantage. If a single disk in the array fails, the entire data set becomes inaccessible due to the fragmented nature of data across disks.
- Mean-time to failure is, in fact, less than standard disks
- No parallelism for small requests

RAID Levels

RAID Level 1 (Disk mirroring)

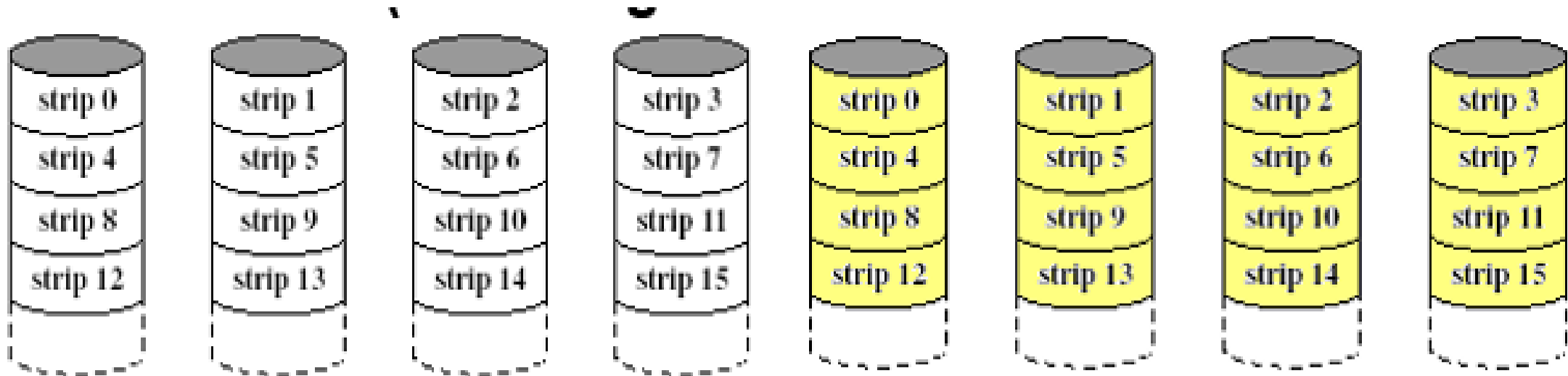
- Also known as **disk mirroring**, this configuration consists of **at least two drives (physical disks) that duplicate the storage of data** - ensuring that if one disk fails, the data can still be accessed from the other disk. This redundancy enhances data reliability and availability, reducing the risk of data loss due to disk failures.
- There is **no striping**.
- **Achieves redundancy** by mirroring all data across multiple disks.
- **Read performance is improved**, since either disk can be read at the same time.
- **Write performance is the same** as for **single disk storage**.



RAID Levels

RAID Level 1

- **Read request** can be serviced from either of available 2 disks, which ever involves the **minimum seek** time and rotational latency
- **Write request** requires both disks to be updated – but this can be done in **parallel**. (Slower write dictates overall speed).
- **Recover from failure** is simple! (data may still be accessed from the second drive)



RAID Levels

RAID Level 1 – Advantages

- **Fast Reads:** Reads can be serviced from either disk, whichever is faster at that moment. This can improve read performance compared to a single disk.
- **Easy Recovery:** If a disk fails, the system simply uses the surviving mirror to access data. No complex rebuild process is needed.

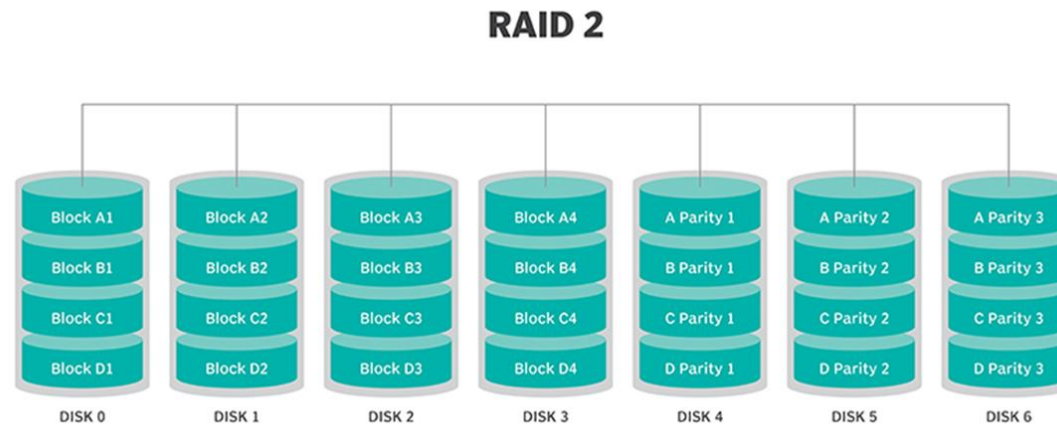
Disadvantages

- **Cost:** Requires double the storage compared to a single disk for the same usable capacity.
- **Limited Capacity:** Usable storage is limited to the size of the smallest disk in the mirror.
- **Limited Use:** Primarily used for critical data due to storage inefficiency.

RAID Levels

RAID Level 2 - Memory-style error correcting code organization (Bit-Level Striping)

- This configuration uses **striping across disks**, with **some disks storing error checking and correcting (ECC) information**.
- RAID 2 also **uses a dedicated Hamming Code parity**, a linear form of ECC
- Called Uses **bit-level striping**
- Some of the bits used for parity or other error correction
- **Sample:** *4-bits of data encoded by additional 3-bits of ECC, and resulting 7 bits stored on 7 different disks*



RAID Levels

- **RAID Level 2 - Advantages**

- Great parallelism for read as well as write
- Even if one disk crashes, all data can be recovered (due to ECC)

- **Disadvantages**

- Drives have to be synchronized
- Overhead in putting together reads from different disks
- Crash recovery much slower (compared to RAID-1)

RAID Levels

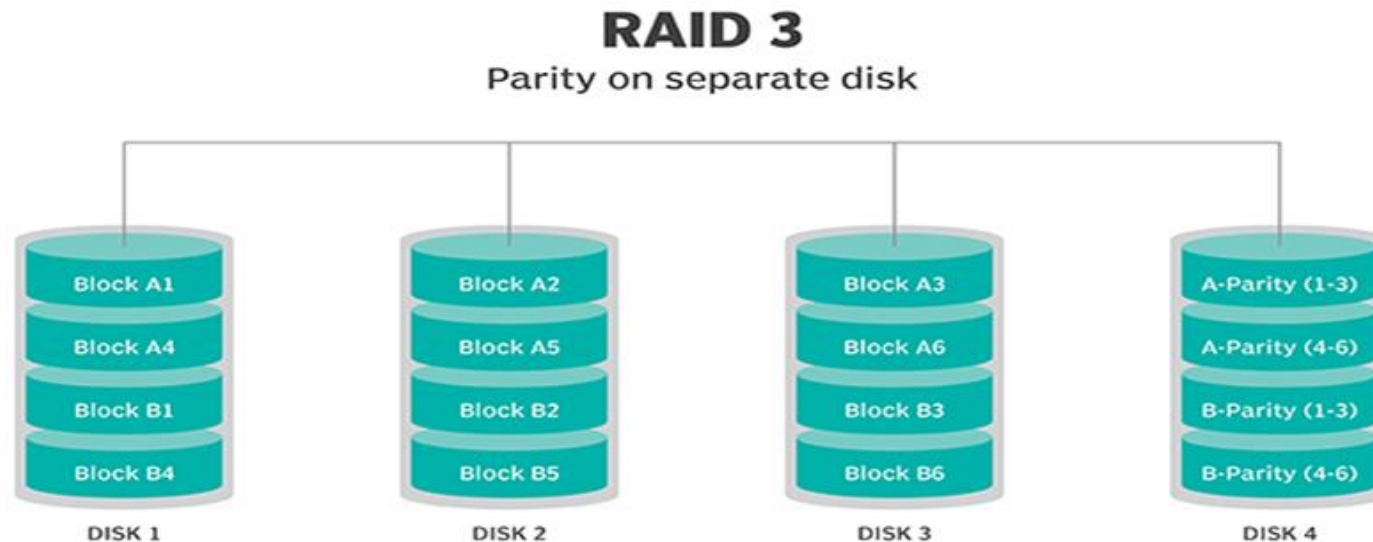
RAID Level 3 - (Byte-Level Striping)

- Requires N+1 disk drives
 - N drives contain data (1/N of each data block)
 - Block $b[k]$ now partitioned into N fragments $b[k,1]$, $b[k,2]$, ... $b[k,N]$
 - Parity drive contains exclusive or of these N fragments
$$p[k] = b[k,1] \oplus b[k,2] \oplus \dots \oplus b[k,N]$$
 - This technique uses striping and dedicates one drive to storing parity information.
 - The embedded ECC information is used to detect errors.
 - Data recovery is accomplished by calculating the exclusive information recorded on the other drives.
 - Because an I/O operation addresses all the drives at the same time, RAID 3 cannot overlap I/O.
 - For this reason, RAID 3 is best for single-user systems with long record applications.

RAID Levels

RAID Level 3

- Each read/write involves all disks in RAID array
 - Cannot do two or more reads/writes in parallel
 - Performance of array not better than that of a single disk



RAID Levels

RAID Level 3 – How Parity Works?

- Truth table for XOR (same as parity)

A	B	$A \oplus B$
0	0	0
0	1	1
1	0	1
1	1	0

D0	D1	P
0	0	0
0	1	1
1	0	1
1	1	0

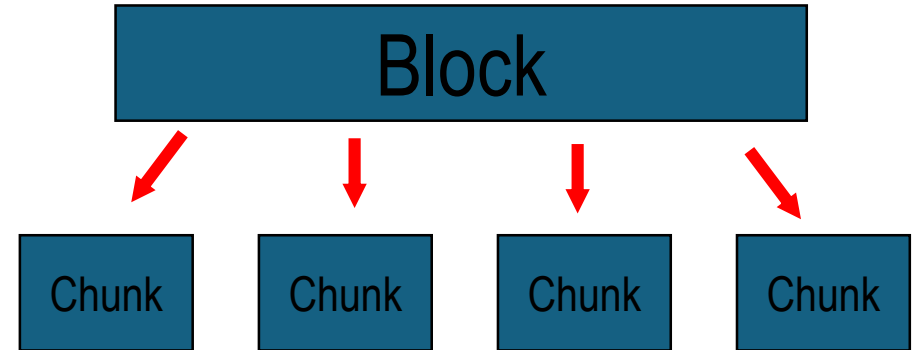
- Recovering from a disk failure
 - Small RAID level 3 array with data disks D0 and D1 and parity disk P can tolerate failure of either D0 or D1

$D1 \oplus P = D0$	$D0 \oplus P = D1$
0	0
0	1
1	0
1	1

RAID Levels

RAID Level 3 – How it Works?

- Assume we have $N + 1$ disks
- Each block is partitioned into N equal chunks

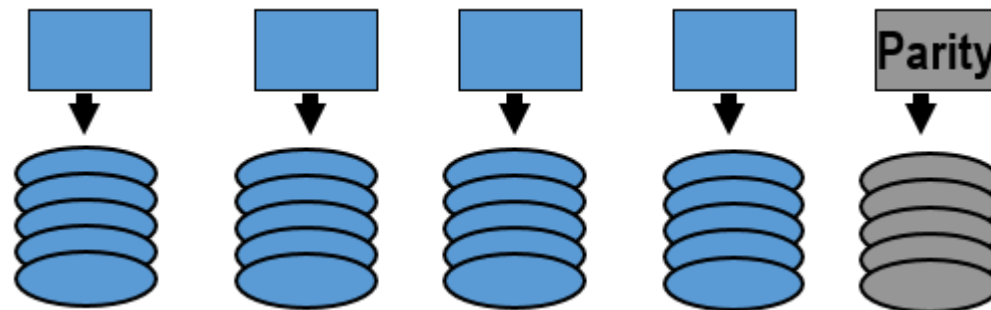


- XOR data chunks to compute the parity chunk

$N = 4$ in example



- Each chunk is written into a separate disk



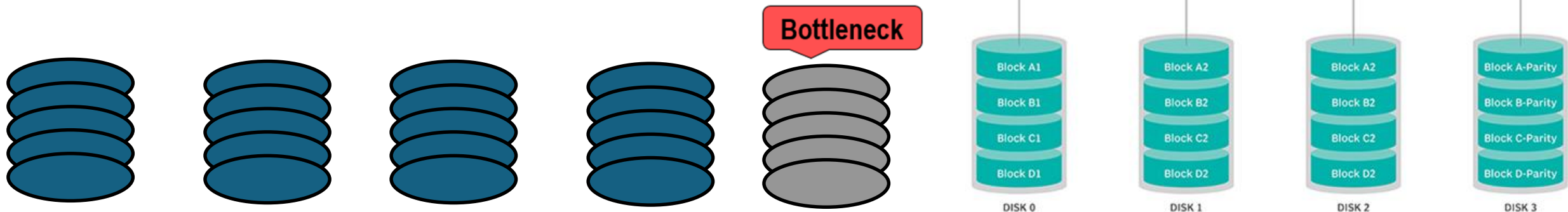
RAID Levels

RAID Level 4 (Block-Level Stripping)

- Requires N+1 disk drives
 - N drives contain data
 - Individual blocks, not chunks
 - Blocks with same disk address form a **stripe**
- Parity drive contains **exclusive or** of the *N* blocks in stripe

$$p[k] = b[k] \oplus b[k+1] \oplus \dots \oplus b[k+N-1]$$

- Parity block now reflects contents of several blocks!
- Can now do parallel reads/writes

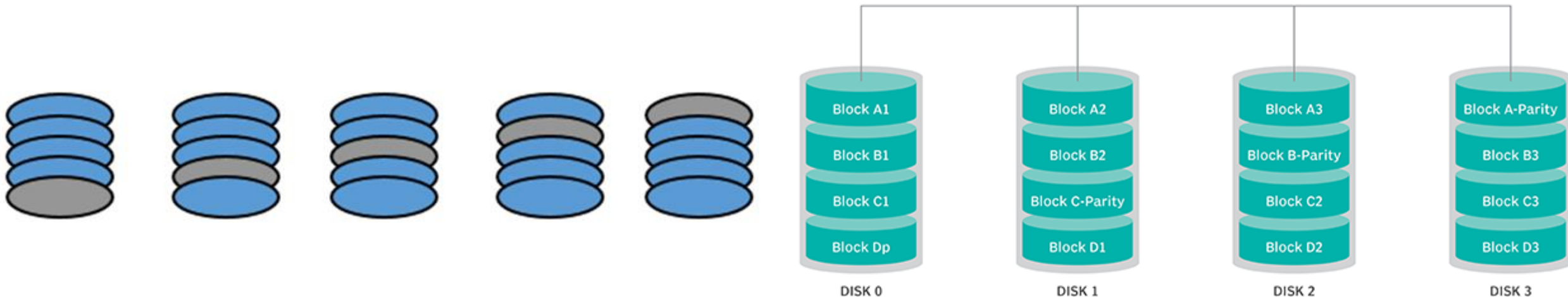


RAID Levels

RAID Level 5

- **Single parity drive** of RAID level 4 is involved in every write
 - *Will limit parallelism*
- RAID-5 **distribute the parity blocks among the N+1 drives**
 - *Much better*

RAID 5



RAID Levels

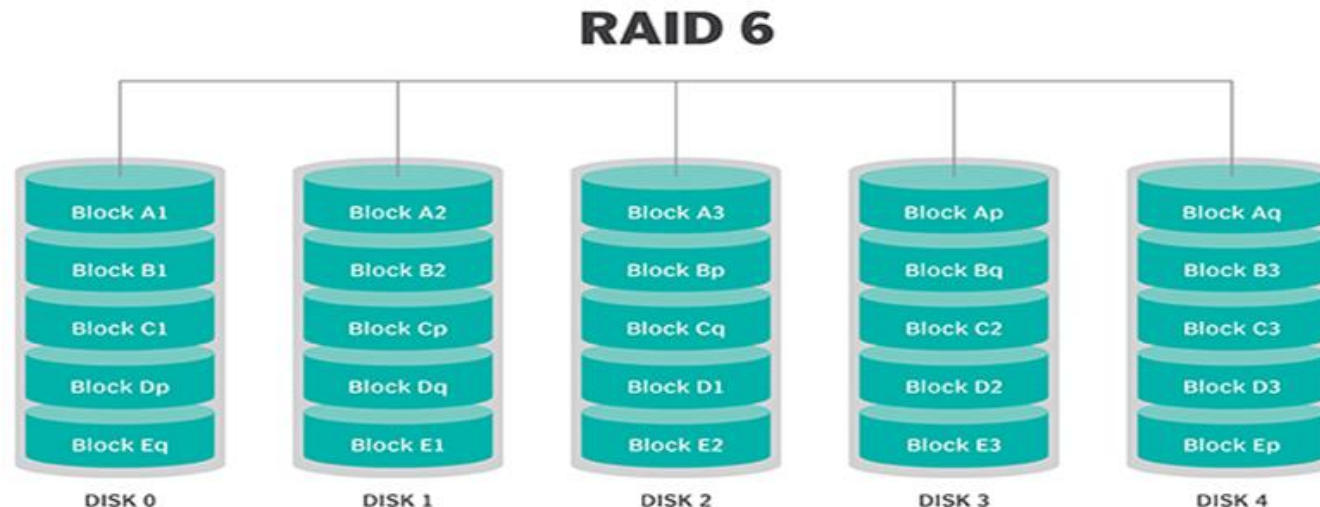
RAID Level 5 (Striping with Parity)

- **Method:** Data is striped across multiple disks with parity information distributed among them.
- **Purpose:** Balances performance, storage efficiency, and fault tolerance.
-  **Advantages:**
 - Fault tolerance — can recover from the failure of one disk using parity data.
 - Efficient storage — only one disk worth of space is lost to parity.
 - Good read speed due to striping.
-  **Disadvantages:**
 - Write performance is slower due to parity calculation.
 - Rebuilding a failed disk can be time-consuming and risky.

RAID Levels

RAID Level 6

- This technique is **similar to RAID 5** - but it includes a **second parity scheme** distributed across the drives in the array.
- The use of additional parity **enables the array to continue functioning, even if two disks fail simultaneously.**
- However, this extra protection comes at a **cost.**
- RAID 6 arrays often have **slower write performance** than RAID 5 arrays.



RAID Levels

RAID Level 6 (Striping with Double Parity)

- Similar to RAID 5, but with two parity blocks for fault tolerance.
- **Purpose:** Enhanced fault tolerance over RAID 5.
-  **Advantages:**
 - Can recover from the failure of two disks.
 - Good read speed.
-  **Disadvantages:**
 - Reduced write speed due to double parity calculation.
 - More storage capacity lost to parity (equivalent to two disks).

Key Differences Between RAID Levels

- RAID 0 – **Maximum performance**, but **no data protection**.
- RAID 1 – **High redundancy** but **lower storage capacity**.
- RAID 2 – Uses ECC, **rarely used** due to **complexity**.
- RAID 3 – **Byte-level striping** with parity on a single disk, **bottleneck on parity disk**.
- RAID 4 – Similar to RAID 3 but with **block-level striping**; **parity disk** creates a bottleneck.
- RAID 5 – **Balanced performance, redundancy, and storage efficiency**.
- RAID 6 – **Higher fault tolerance** than RAID 5 at the **cost** of performance.
- RAID 10 – Combines the performance of RAID 0 with the redundancy of RAID 1.

RAID Level	Striping	Parity	Fault Tolerance	Minimum Disks	Read Speed	Write Speed	Storage Efficiency	Best Use Case
RAID 0	✔ Yes	✗ No	✗ None	2	🚀 Very High	🚀 Very High	100%	High-speed data access where fault tolerance is not needed
RAID 1	✗ No	✗ No	✔ 1 disk	2	🚀 High	📄 Moderate	50%	Critical data needing redundancy
RAID 2	✔ Yes	✔ ECC (Error Correction Code)	✔ Multiple disk failure (depends on ECC)	3+	🚀 High	✗ Slow	Low (depends on ECC)	Rarely used due to complexity
RAID 3	✔ Byte-level	✔ Single disk parity	✔ 1 disk	3+	🚀 High	📄 Slow	$(n-1)/n$	Video editing, large files

RAID Level	Striping	Parity	Fault Tolerance	Minimum Disks	Read Speed	Write Speed	Storage Efficiency	Best Use Case
RAID 4	 Block-level	 Dedicated parity disk	 1 disk	3+	 High	 Slow (parity bottleneck)	$(n-1)/n$	Large sequential data sets
RAID 5	 Block-level	 Distributed parity	 1 disk	3+	 High	 Moderate	$(n-1)/n$	General-purpose storage, web servers
RAID 6	 Block-level	 Double parity	 2 disks	4+	 High	 Slow (parity calculation)	$(n-2)/n$	Mission-critical storage, database servers
RAID 10	 Yes	 No	 Multiple disk failures (if mirrored pairs remain intact)	4+ 	 Very High	 High	50%	High-performance applications needing fault tolerance

Reliability of Memory Systems

- **Fault tolerance for Disks**

- Disks can have lots of errors
 - **Corrupted bits**
 - **Corrupted sectors**
 - Manufacturing defects
 - Disk crashes
- Many levels of protection implemented
 - **Error correction within sectors**
 - Extra sectors within a disk
 - Mirroring
 - RAID (redundancy and parallelism)

Error Detecting and Error Correcting Systems

- Error detection and correction code plays an important role in the storing and fetching data in the secondary device.
- The bits of the data may change(either 0 to 1 or 1 to 0) during storage and retrieval.
- But it is possible to get back the original data.
- For this purpose, we first need to detect either an error is present or not using error detection codes.
- If the error is present in the code, then we will correct it with the help of error correction codes.

Message Format

- A typical disk sector occupies 512 bytes.
 - **Preamble** identifying start code and sector address.
 - **Data**
 - **Error correction code (16 bits)** - At least detecting errors possible with probability almost 1
 - ECC – generated from the original data bits

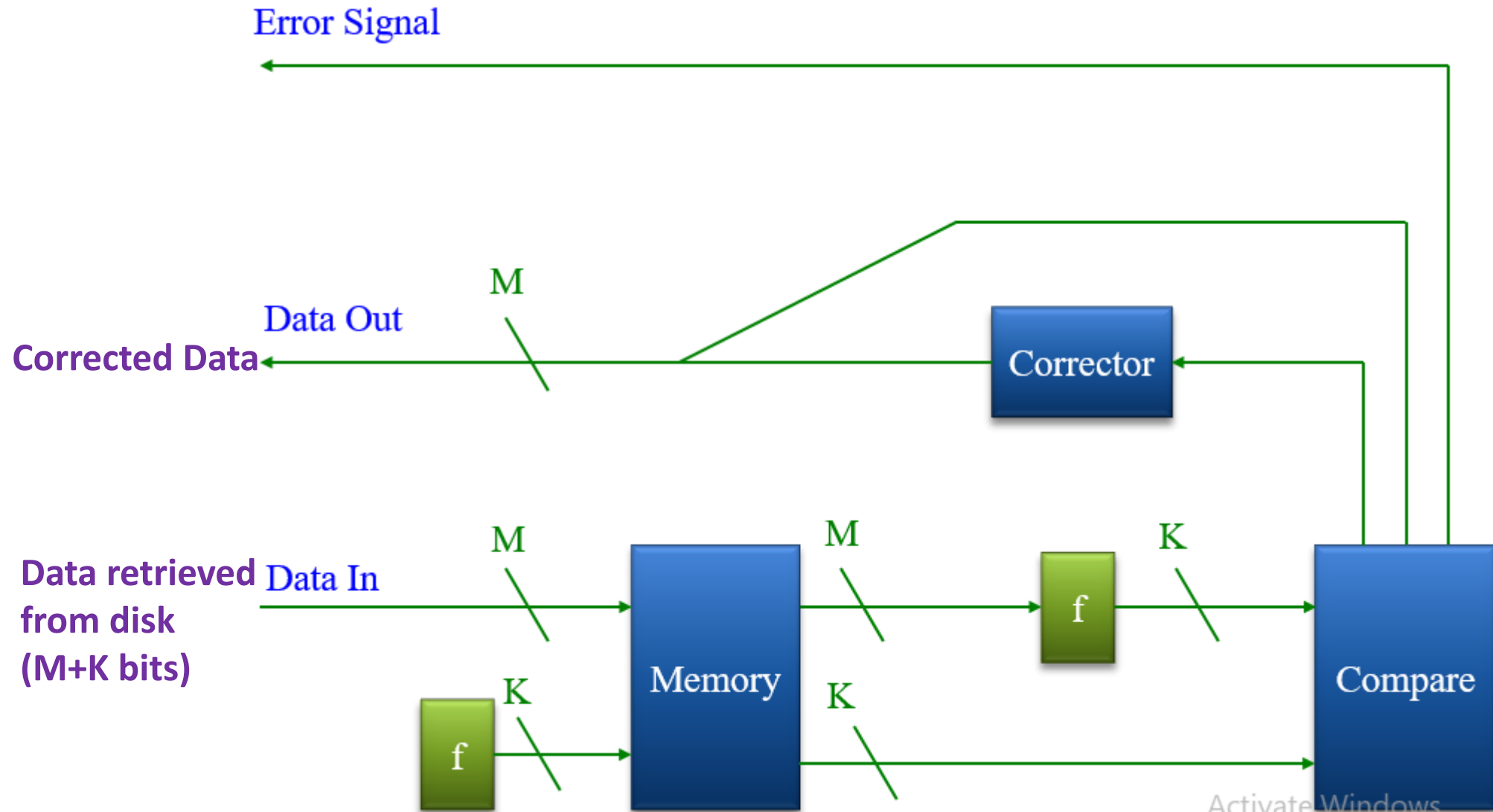


A disk sector

Error Correction Technique

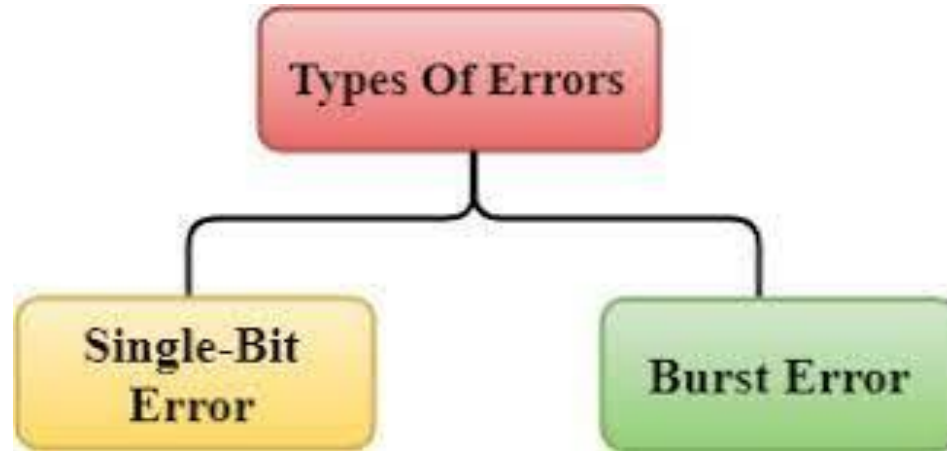
- Basic technique
 - Prior to storing data, a code is generated from the bits in the word.
 - Code stored alongside the word in memory.
 - Code used to identify and correct errors.
 - When the word is fetched a new code is generated and compared to the stored code.
 - No error (normal case)
 - Correctable error is detected and corrected.
 - Non-fixable error is detected and reported.

Error Correction Function

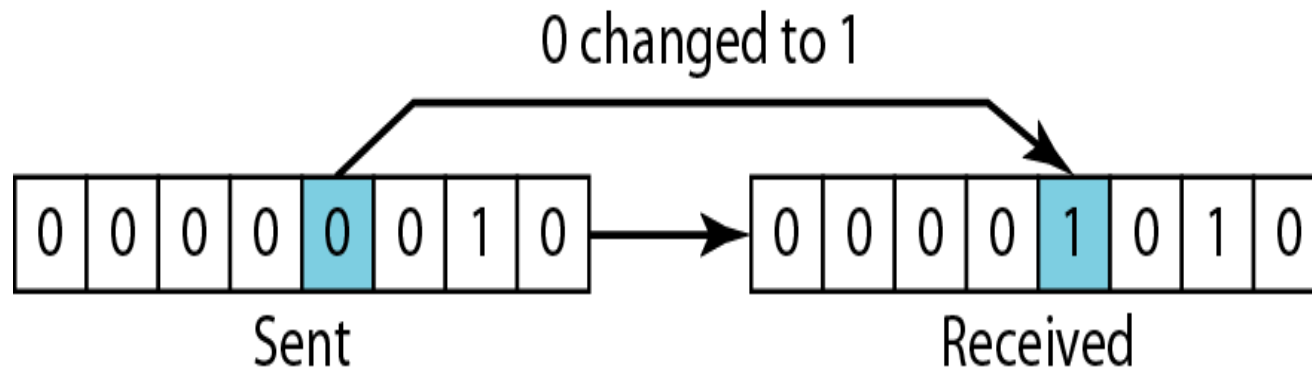


Error Detecting and Error Correcting Systems

Single-bit error



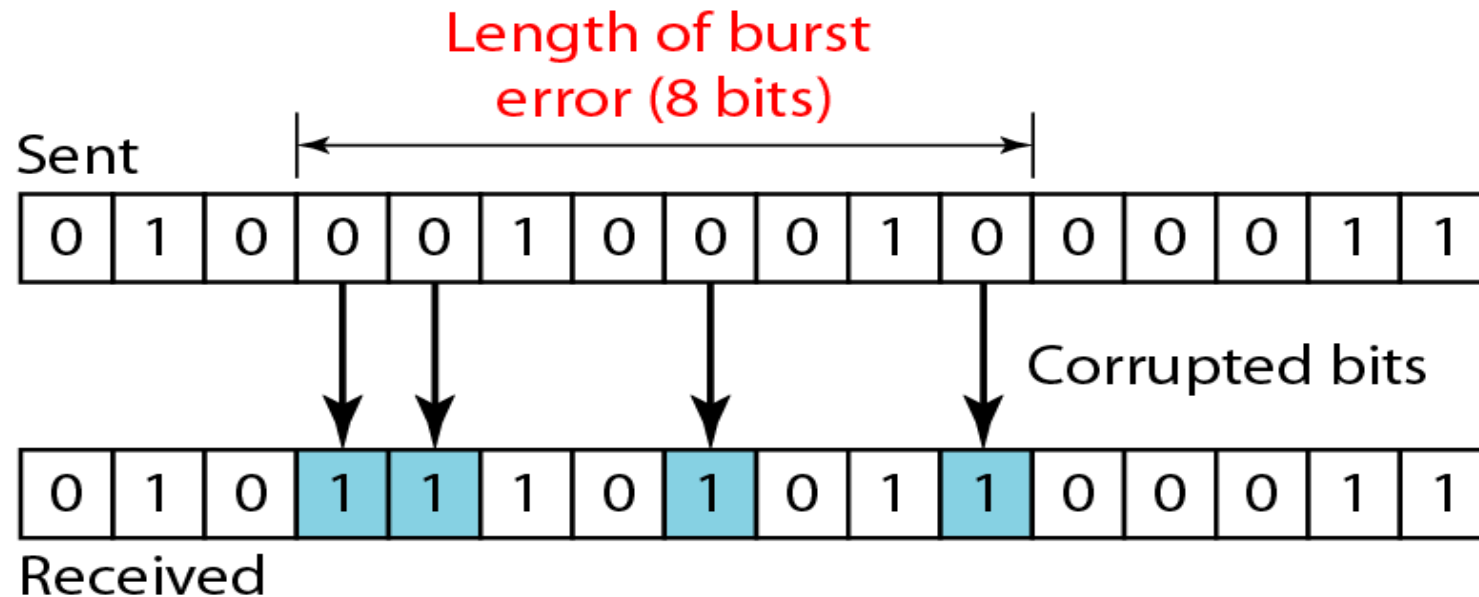
In a single-bit error, only 1 bit in the data unit has changed.



Error Detecting and Error Correcting Systems

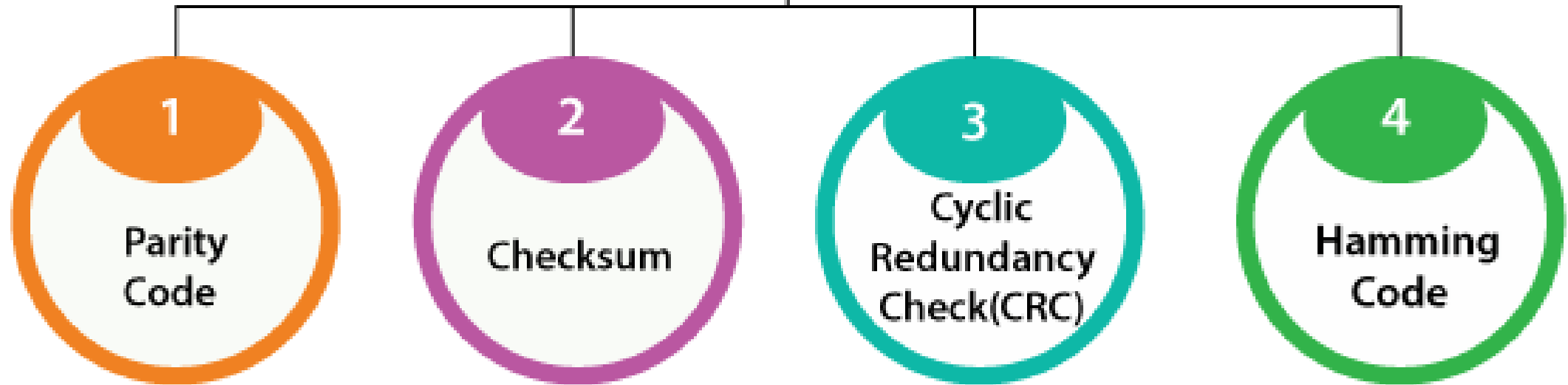
Burst error of length 8

A burst error means that 2 or more bits in the data unit have changed.



Error Detecting and Error Correcting Systems

Error Detection and Correction Techniques



Error Detecting and Error Correcting Systems

To detect or correct errors, we need to send extra (redundant) bits with data.

Error detection codes – are used to detect the errors present in the received data bitstream. These codes contain some bits, which are appended to the original bit stream. These codes detect the error, if it is occurred during transmission of the original data

Example – Parity code, Hamming code.

Error correction codes – are used to correct the errors present in the received data bitstream so that, we will get the original data. Error correction codes also use the similar strategy of error detection codes.

Example – Hamming code.

Error Detecting and Error Correcting Systems

Parity Code

- It is easy to **append one parity bit either to the left of MSB or to the right of LSB** of original bit stream. There are two types of parity codes, namely even parity code and odd parity code based on the type of parity being chosen.

Even Parity Code

- The **value of even parity bit should be zero**, if **even number of ones present** in the binary code. Otherwise, **it should be one**. So that, even number of ones present in even parity code. Even parity code contains the data bits and even parity bit.

Binary Code	Even Parity bit	Even Parity Code
000	0	0000
001	1	0011
010	1	0101
011	0	0110
100	1	1001
101	0	1010
110	0	1100
111	1	1111

Error Detecting and Error Correcting Systems

Odd Parity Code

- The value of odd parity bit should be zero, if odd number of ones present in the binary code. Otherwise, it should be one. So that, odd number of ones present in odd parity code. Odd parity code contains the data bits and odd parity bit.

Binary Code	Odd Parity bit	Odd Parity Code
000	1	0001
001	0	0010
010	0	0100
011	1	0111
100	0	1000
101	1	1011
110	1	1101
111	0	1110

Error Detecting and Error Correcting Systems

Hamming Code - working

- Most common types of error-correcting codes used in RAM
- devised by R. W. Hamming.
- In the Hamming code, **k parity bits** are added to an **n-bit data word**, forming a new word of **n + k bits**.
- The bit **positions** are numbered in sequence from **1 to n + k**.
- **Positions** numbered with **powers of two** are reserved for the parity bits. ($2^0, 2^1, 2^2, 2^3, 2^4..$)
- **Remaining** bits are the **data** bits.

Error Detecting and Error Correcting Systems

Hamming Code – Data Storage with data and parity bits

Problem 1: (Data Storage)

Let us assume that the original data to be stored is 1110. Find the number of parity bits and the encoded data (actual data + parity bits) stored in the disk.

1. Number of parity bit calculation

$2^r \geq m + r + 1$ where, r = redundant bit (or) parity bit, m = data bit

Example: Consider hamming code of data is 1110 and number of data bits (m) = 4, calculate number of parity bits.

If $r=1 \rightarrow 2^1 \geq 4 + 1 + 1 = 2 \geq 6 \rightarrow$ equation is not satisfied-**wrong** assumption

If $r=2 \rightarrow 2^2 \geq 4 + 2 + 1 = 4 \geq 7 \rightarrow$ equation is not satisfied-**wrong** assumption

If $r=3 \rightarrow 2^3 \geq 4 + 3 + 1 = 8 \geq 8 \rightarrow$ equation is satisfied-**correct** assumption

Result is if m (data bits) is 4 and r (parity bits) is 3. Total bits=7

Error Detecting and Error Correcting Systems

Hamming Code – Data Storage with data and parity bits

Problem 1: (Data Storage)

Let us assume that the original data to be stored is 1110. Find the number of parity bits and the encoded data (actual data + parity bits) stored in the disk.

2. Placement of Parity Bits

Power of Two	2^0	2^1		2^2			
Bit Location	1	2	3	4	5	6	7
Bits	P1	P2	M1	P3	M2	M3	M4

Placement of Parity Bits

P1, P2, and P3 are parity bits.

M1, M2, M3, and M4 are message bits

Error Detecting and Error Correcting Systems

Hamming Code – Data Storage with data and parity bits

Problem 1: (Data Storage)

Let us assume that the original data to be stored is 1110. Find the number of parity bits and the encoded data (actual data + parity bits) stored in the disk.

3. Hamming Code with Even Parity

- Let us determine Hamming code for the message to be transmitted **1110**.
- Once we get the message to be transmitted, we can fix the places for **M1**, **M2**, **M3**, and **M4**, as shown below.

Power of Two	2^0	2^1		2^2			
Bit Location	1	2	3	4	5	6	7
Bits	P1	P2	1	P3	1	1	0

Even Bit Parity

Error Detecting and Error Correcting Systems

Hamming Code – Data Storage with data and parity bits

4. Rules to Determine the Parity Sequence

	P3	P2	P1	Decimal Equivalent
	0	0	0	0
P1(1,3,5,7) ←	0	0	1	1
P2(2,3,6,7)	0	1	0	2
P3(4,5,6,7)	0	1	1	3
	1	0	0	4
	1	0	1	5
	1	1	0	6
	1	1	1	7

$P1=(1,3,5,7) \rightarrow (P1,1,1,0) \rightarrow$ Even number of 1's $\rightarrow P1=0$

$P2=(2,3,6,7) \rightarrow (P2,1,1,0) \rightarrow$ Even number of 1's $\rightarrow P2=0$

$P3=(4,5,6,7) \rightarrow (P3,1,1,0) \rightarrow$ Even number of 1's $\rightarrow P3=0$

Power of Two	2^0	2^1	2^2			
Bit Location	1	2	3	4	5	6
Bits	P1	P2	1	P3	1	1

So, the encoded data is:
0010110

Error Detecting and Error Correcting Systems

Hamming Code – Data Retrieval, Error Detection and Correction

Problem 2: (Case 1 – Error detection)

Let us assume that the data retrieved in (7,4) format is 1110000. Check the error with even parity bit.

(i) Parity bits = 3

Power of Two	2^0	2^1		2^2			
Bit Location	1	2	3	4	5	6	7
Bits	P1	P2	M1	P3	M2	M3	M4

Placement of Parity Bits

Error Detecting and Error Correcting Systems

Hamming Code – Data Retrieval, Error Detection and Correction

Consider the following table.

(ii)

Location	1	2	3	4	5	6	7
Bits	P1	P2	M1	P3	M2	M3	M4
P1(1,3,5,7)	1	1	1	0	0	0	0
P2(2,3,6,7)							
P3(4,5,6,7)							

Hamming Code Error Detection

(iii)

Error					Even Parity
E1(1,3,5,7)	1	1	0	0	E1=0
E2(2,3,6,7)	1	1	0	0	E2=0
E3(4,5,6,7)	0	0	0	0	E3=0
Error Code	(E3E2E1)=(000)				

Hamming Code Error Detection Bits

The resultant error is now (000), which means that **no error has been found in transmitting the data**

Error Detecting and Error Correcting Systems

Hamming Code – Data Retrieval, Error Detection and Correction

Problem 2: (Case 2 – Error detection and Correction)

Let us assume that the data retrieved in (7,4) format is 1111000. Check the error with even parity bit.

(i) Parity bits = 3

Power of Two	2^0	2^1		2^2			
Bit Location	1	2	3	4	5	6	7
Bits	P1	P2	M1	P3	M2	M3	M4

Placement of Parity Bits

Error Detecting and Error Correcting Systems

Hamming Code – Data Retrieval, Error Detection and Correction

Consider the following table.

(ii)

Location	1	2	3	4	5	6	7
Bits	P1	P2	M1	P3	M2	M3	M4
P1(1,3,5,7)	1	1	1	1	0	0	0
P2(2,3,6,7)							
P3(4,5,6,7)							

Hamming Code Error Detection

(iii)

Error					Even Parity
E1(1,3,5,7)	1	1	0	0	E1= 0
E2(2,3,6,7)	1	1	0	0	E2= 0
E3(4,5,6,7)	1	0	0	0	E3= 1
Error Code	(E3E2E1)=(100)				

Hamming Code Error Detection Bits

The resultant error is now (100) → decimal value is 4 so **Error in bit 4 → Error Detection**
Error Correction is → the value of bit 4 is 1 → change the value to 0
now the answer is 1110000

Hamming Code - Problems

- Suppose an 8-bit data word stored in memory is 11000010. Using the Hamming algorithm, determine what check bits would be stored in memory with the data word. Show how you got your answer.
- For the 8-bit word 00111001, the check bits stored with it would be 0111. Suppose when the word is read from memory, the check bits are calculated to be 1101. What is the data word that was read from memory?
- How many check bits are needed if the hamming error correction code is used to detect single bit errors in 1024 bit data word
- Generate the code for the data word 0101000000111001. show that the code will correctly identify an error in data bit 5.

I/O Performance

Performance Metrics:

- **seek time:** move the arm assembly to track. The time taken to move that Read/Write head(RWH) from any place to the required track is called as seek time.
- **Rotational delay:** wait for sector to come around. Period of time it takes for the desired sector to come under the RWH.
- **transfer time:** get the bits off the disk
- **Controller time:** time for setup

Average access time (AVGAT) = SeekTime + Rotational Latency + Data Transfer Time + Control Delay (i.e., Delay used to setup)

I/O Performance

Example:

- Average time to read/write 512-byte sector
 - Disk rotation at 10,000 RPM
 - Seek time: 6ms
 - Transfer rate: 50 MB/sec
 - Controller overhead: 0.2 ms
- Average time:
 - Seek time + rotational delay + transfer time + controller overhead
 - $6\text{ms} + 0.5 \text{ rotation} / (10,000 \text{ RPM}) + 0.5\text{KB} / (50 \text{ MB/sec}) + 0.2\text{ms}$
 - $6.0 + 3.0 + 0.01 + 0.2 = 9.2\text{ms}$